Cobaya Tutorial

A Study Note of Cosmology

Aofei Sang, Xue Zhou[1]

Department of Physics, Southern University of Science and Technology

Shenzhen, China

2026.06.v1

[1] 12332935@mail.sustech.edu.cn

Foreword

This tutorial is a summary of our coursework and research experience in modern cosmology, aiming to offer a hands-on, theory-integrated guide to three primary numerical tools: `CAMB`, a Boltzmann solver for computing theoretical cosmological observables; `Cobaya`, a Bayesian inference framework for parameter estimation and model comparison; and `GetDist`, a standard package for analysis and visualization of MCMC posterior samples.

The full PDF of this tutorial, alongside supplementary example files and demonstration Jupyter notebooks, is openly hosted, maintained, and updated in our GitHub repository: `https://github.com/XueZHOU-cosmology/Tutorial`.

While we have made every effort to ensure the accuracy of the content and the completeness of its foundational framework, this work can only capture a small fraction of the vast and rich field of cosmology. Owing to the limited scope of this note, many topics cannot be explored in the depth they merit. We sincerely welcome feedback, corrections, and suggestions, and will promptly address any issues raised in future updates.

We hope that this tutorial may help you to explore the fundamental questions of the universe with greater confidence.

Structure of the Note

In Chapter 1, we cover the installation and configuration of the numerical environment, including Python, `Cobaya`, `CAMB`, and cosmological datasets and likelihoods, alongside a test run to validate the full setup. In Chapter 2, we review the theoretical framework of background cosmology and linear cosmological perturbation theory. In Chapter 3, we introduce the core cosmological observables linking theoretical predictions to observations. Finally, in Chapter 4, we detail the implementation of the `CAMB` code and provide practical examples for MCMC result analysis and visualization using `Cobaya` and `GetDist`.

Contents

Chapter 1

Software Preparation

This chapter provides a comprehensive guide to setting up the software environment required for the cosmological analyses presented in this tutorial. We cover the configuration of the Python development environment, followed by the installation and basic usage of the three core tools: `Cobaya`, `CAMB`, and cosmological datasets and likelihoods.

Section 1.1 is intended for users who are new to Python and wish to set up a development environment using either Anaconda Navigator or Visual Studio Code (VS Code). Both platforms are user-friendly and widely used; readers only need to follow the instructions for their preferred platform.

If you already have a Python environment configured, you may proceed directly to Section 1.2, which introduces the installation and basic usage of `Cobaya`, and performs a toy running with the default setup to ensure everything is ready.

All commands described here are run under the macOS system.

1.1 Code Editing and Execution Environment

Following these steps, you can download and install the full version of Anaconda, which includes conda (environment/package manager), the latest Python, pip (Python package installer), data science packages (NumPy, pandas, Matplotlib, scikit-learn, etc.) and Anaconda Navigator (GUI for managing environments and packages) on your computer:

- Visit the official website https://www.anaconda.com and click [Free Download].
- Select the installer for your computer system (Windows, macOS, or Linux).
- Choose [Anaconda Distribution] (then [Graphical Installer] for macOS). The [Command Line] is only recommended under the following circumstances: (a) you are operating on a remote server or in an environment without a graphical interface (uncommon for macOS); (b) you need to deploy Anaconda in a batch manner using

an automated script; (c) you are accustomed to controlling the installation process entirely through the terminal.

- After finishing the download, follow the prompts to install. During the installation, please keep the default installation path (`/home/[Your Username]/anaconda3` for Linux, and `/Users/[Your Username]/anaconda3` for macOS) and type [yes] when asked: [Do you wish to initialize Anaconda3 by running conda init?] (otherwise, the `conda` command will not work).

- Verify the installation. Open a terminal window and run:

```
conda --version
# Example output: conda 24.9.0
python --version
# Example output: Python 3.13.5
pip --version
# Example output:
#pip 25.3 from
    .../anaconda3/lib/python3.13/site-packages/pip
python -c "import numpy; print('NumPy version:',
    numpy.__version__)"
# Example output: NumPy version: 1.26.4
```

- The prerequisites for `Cobaya` are Python version ≥ 3.8 and the Python package manager pip version ≥ 20.0.

If you see version numbers, the installation is successful! Should you encounter any issues during this process, you may seek assistance from someone familiar with the procedure or consult an AI.

After setting up Python, conda, and pip on your computer, you will need a code editor to write and execute your code. Here are two recommendations to suit different workflow needs: [Anaconda Navigator] (supports Python and R) is designed for visualization, data science, and machine learning. It features a user-friendly graphical interface and efficiently manages independent environments for different projects through conda; [Visual Studio Code] (supports multiple languages: Python, R, C, C++, Java, LaTeX, and more) is suitable for cross-language development and boasts a rich ecosystem of extensions.

1.1.1 Anaconda Navigator

After finishing the Anaconda installation above, you will find the [Anaconda-Navigator] on your desktop.

You can create distinct projects/environments and manage their respective package dependencies on the [Environments] tab. On the [Home] tab, you can select a specific project environment and execute corresponding operations using the applications displayed here. The following three applications are commonly used:

- ☐ [Spyder]: Spyder enables the execution of [.py] files with an intuitive interface design. The [Variable Explorer] panel allows tracking of variable values, making this tool highly efficient for processing complex data and creating data visualizations. Additionally, Spyder can serve as an editor for `.f90`, `.yaml`, and other file types. Its official tutorials are available at: https://docs.spyder-ide.org/current/index.html.

- ☐ [JupyterLab] and [Jupyter Notebook]: These tools support running [.ipynb] files, and are widely used for creating and disseminating tutorials, example documents, and technical literature across various fields. Their tutorials are available at: https://jupyterlab.readthedocs.io/en/stable/index.html.

1.1.2 Visual Studio Code

If you prefer an execution environment that supports multiple languages with a rich ecosystem of extensions, you can download VS Code from https://code.visualstudio.com:

- Install VS Code by following the on-screen prompts, and check the option [Add to PATH] during installation for easy access via the terminal.
- Open VS Code, click the [Extensions] icon (the square-shaped icon in the left sidebar), search for and install the [Python] extension published by Microsoft.
- Click the [File] tab in the top-left corner to open an existing [.py] or [.ipynb] file, or select [New File] to create a new one. In the bottom-right corner of the VS Code interface, click the Python environment indicator (showing the current interpreter name) to select an existing Python environment or create a new environment, and confirm the programming language you are working with.

There are two ways to manage different Python development environments in VS Code:

- Directly open the target project folder (via [File > Open Folder]), open the integrated terminal corresponding to the folder (by right-clicking and selecting New Terminal at Folder), and use pip/conda commands to install/uninstall packages and create/remove environments;
- Manage environments in conjunction with Anaconda Navigator, link your conda environments to VS Code by selecting the conda interpreter path in the environment selector.

1.2 Cobaya, Data, and Boltzmann Solver Installations

Throughout this section, we will follow the instructions provided on the website `https://cobaya.readthedocs.io/en/latest/`, which contains rich and detailed information about Cobaya.

1.2.1 Installations

To install the cosmological inference code `Cobaya`, the Boltzmann solvers `CAMB` and `CLASS`, as well as standard cosmological datasets and likelihoods (including Planck, BAO, and SNe Ia), please follow the step-by-step instructions provided in the official documentation: `https://cobaya.readthedocs.io/en/latest/installation.html#` and `https://cobaya.readthedocs.io/en/latest/installation_cosmo.html`.

For the South Pole Telescope (SPT) CMB data and its official likelihood implementation, full tutorials and usage examples are provided in the dedicated repository: `https://github.com/SouthPoleTelescope/spt_candl_data?tab=readme-ov-file`.

1.2.2 Toy Run: ΛCDM Model and CPL Model

To ensure that all packages function properly, you can perform a quick basic cosmology run with default `CAMB` versions by following the tutorial at `https://cobaya.readthedocs.io/en/latest/cosmo_basic_runs.html`.

For example, first create a prior input file ([.yaml] or Python format), which can be generated by executing the following command:

```
cobaya cosmo-generator
```

Note: The local `generator` GUI requires the `PySide6` package, which can be installed via

```
pip install PySide6
```

If you do not wish to install additional dependencies, use the web-based Streamlit version at `https://cobaya-gui.streamlit.app/`.

Input examples for both Python and [.yaml] files are available in the `example` folder.

Using [.yaml] Files as an Example

You can add the following line to your input [your_input_file_name.yaml] file:

```
output: /path/to/your/chains/folder/chain_root_name
```

and then run Cobaya with the command:

```
cobaya-run [your_input_file_name.yaml]
```

If the sampling begins, the installation and configuration are successful. Otherwise, there may be issues in the environment or setup.

Chapter 2

Theoretical Framework of Cosmology

Modern theoretical cosmology is built on the synthesis of core physical theories — Einstein's general relativity (GR), quantum field theory, particle physics, and statistical physics — and anchored by rigorous observational constraints, establishing a self-consistent framework for describing the origin, evolution, structure formation, and ultimate fate of the Universe.

Robustly confirmed by all-sky cosmic microwave background (CMB) surveys, galaxy redshift surveys, and measurements of the diffuse extragalactic background light, the Universe is statistically homogeneous and isotropic on sufficiently large scales (~ 100Mpc)[1–4]. This property, known as the Cosmological Principle, forms the minimal symmetry assumption underpinning modern cosmology; when combined with GR, it fully determines the Universe's expansion history.

On smaller scales, by contrast, the Universe shows pronounced inhomogeneities and anisotropies that break the statistical symmetry of the cosmic mean density field. Yet it is the tiny primordial deviations that seed all cosmic structure observed today. Through gravitational instability, these minute primordial overdensities underwent linear growth in the early Universe, followed by non-linear collapse as cosmic evolution proceeded. This process gave rise to galaxies, galaxy clusters, supermassive black holes, and planetary systems, ultimately weaving the complex cosmic web that defines both our local environment and the Universe's large-scale structure.

This chapter investigates the cosmic expansion dynamics of the homogeneous and isotropic Universe, and describes the evolution of primordial fluctuations and cosmic structure formation, laying the theoretical underpinnings for the subsequent analyses in this work.

2.1 The Homogeneous and Isotropic Universe

Extending the Copernican Principle that we occupy no privileged position in the Universe, the Cosmological Principle is mathematically framed in two core symmetry postulates:

- Spatial homogeneity: At any given cosmic time t, all physical properties of the Universe are statistically indistinguishable for comoving observers at any spatial location on the constant-time hypersurface Σ_t. In other words, there are no preferred spatial positions in the Universe.

- Spatial isotropy: The Universe exhibits identical physical characteristics and observational signatures to any fundamental comoving observer, regardless of the direction of observation. In other words, there are no preferred directions in the Universe.

Crucially, in a universe that is isotropic around every point, homogeneity follows automatically [5–8]. As we will show, these symmetries together with GR determine the Universe's background evolution equations.

2.1.1 The Friedmann-Lemaître-Robertson-Walker Metric

Under the Cosmological Principle, the unique 4-dimensional Lorentzian spacetime metric is the Friedmann-Lemaître-Robertson-Walker (FLRW) metric [8–12]:

$$\mathrm{d}s^2 = g_{\mu\nu}\mathrm{d}x^\mu\mathrm{d}x^\nu = -c^2\mathrm{d}t^2 + a(t)^2\gamma_{ij}\mathrm{d}x^i\mathrm{d}x^j, \tag{2.1}$$

where:

- $g_{\mu\nu} = \mathrm{diag}(-1, a(t)^2, a(t)^2, a(t)^2)$ is the metric tensor, and we work in comoving coordinates with the 4-coordinate defined as $x^\mu = (ct, x^i)$.

- $a(t)$ is the dimensionless cosmic scale factor, normalized to $a(t_0) = 1$ at the present cosmic time t_0;

- γ_{ij} is the spatial metric of a 3-dimensional maximally symmetric space with a constant intrinsic curvature K, depending on our choice of spatial coordinates:

$$\gamma_{ij}\mathrm{d}x^i\mathrm{d}x^j = \frac{\delta_{ij}\mathrm{d}x^i\mathrm{d}x^j}{\left(1+\frac{1}{4}K\rho^2\right)^2}, \tag{2.2}$$

$$\gamma_{ij}\mathrm{d}x^i\mathrm{d}x^j = \mathrm{d}r^2 + \chi^2(r)\left(\mathrm{d}\theta^2 + \sin^2\theta\mathrm{d}\phi^2\right), \tag{2.3}$$

$$\gamma_{ij}\mathrm{d}x^i\mathrm{d}x^j = \frac{\mathrm{d}R^2}{1-KR^2} + R^2\left(\mathrm{d}\theta^2 + \sin^2\theta\mathrm{d}\phi^2\right). \tag{2.4}$$

where in Eq. (2.2):

$$\rho^2 = \sum_{i,j=1}^{3} \delta_{ij} x^i x^j, \quad \text{and} \quad \delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{else,} \end{cases} \tag{2.5}$$

and in Eq. (2.3):

$$\chi(r) = \begin{cases} r & \text{in the Euclidean case,} \quad K = 0, \\ \frac{1}{\sqrt{K}} \sin\left(\sqrt{K} r\right) & \text{in the spherical case,} \quad K > 0, \\ \frac{1}{\sqrt{|K|}} \sinh\left(\sqrt{|K|} r\right) & \text{in the hyperbolic case,} \quad K < 0. \end{cases} \tag{2.6}$$

Note that both r and R above are dimensional comoving radial coordinates with units of length, consistent with our convention of dimensionless $a(t)$.

It is a common convention in introductory textbooks to normalize the scale factor such that $K = \pm 1$ whenever $K \neq 0$. In that convention, both the radial coordinates and the curvature K become dimensionless, while the scale factor $a(t)$ carries the dimension of length. For the flat case $K = 0$, the normalization of $a(t)$ is left arbitrary.

Instead, in this paper we adopt the modern cosmological convention, which keeps $a(t)$ dimensionless with the fixed normalization $a(t_0) = 1$ at the present time. In this convention, the spatial curvature K retains its natural dimension of length^{-2}, and all comoving spatial coordinates carry length dimension. The comoving curvature radius of the universe can be defined as

$$R_c = \frac{1}{\sqrt{|K|}},$$

which is the characteristic comoving length scale at which curvature effects become significant. The dimensionless topological index is defined as $k = \mathrm{sign}(K)$, leading to

$$K = \frac{k}{R_c^2}, \quad k \in \{0, +1, -1\}.$$

2.1.2 Energy-Momentum Tensor

The most general form of the energy-momentum tensor for a homogeneous and isotropic Universe is that of a perfect fluid:

$$T_{\mu\nu} = \left(\rho + \frac{P}{c^2}\right) u_\mu u_\nu + P g_{\mu\nu}, \tag{2.7}$$

where ρ is the total energy density of the cosmic fluid, and P is the isotropic pressure of the fluid. Both quantities depend only on the cosmic time t, consistent with the homogeneity and isotropy of spacetime.

In the comoving frame, the 4-velocity of the cosmic fluid is defined as $u^\mu \equiv \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}$, where τ is the proper time of the fluid element. For comoving observers, the proper time coincides with the cosmic time t, such that $u^\mu = c\delta^\mu_0$, which satisfies the standard normalization condition $u^\mu u_\mu = -c^2$. Thus the energy-momentum tensor (2.7) reads:

$$T^\mu_\nu = \mathrm{diag}\left(-\rho c^2, P, P, P\right). \tag{2.8}$$

2.1.3 Friedmann and Continuity equations

To describe the background dynamics of the Universe, Einstein's field equations relate the curvature of spacetime to its energy-momentum content:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}, \tag{2.9}$$

where the Einstein tensor is defined as

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}, \tag{2.10}$$

with $R_{\mu\nu}$ the Ricci tensor and R the Ricci scalar.

For the FLRW metric, the non-zero components of the Einstein tensor are:

$$G_{00} = \frac{3}{c^2}\left(\frac{\dot{a}^2 + Kc^2}{a^2}\right), \tag{2.11}$$

$$G_{ij} = -\frac{1}{c^2}\left(\frac{2\ddot{a}a + \dot{a}^2 + Kc^2}{a^2}\right) a^2 \gamma_{ij}. \tag{2.12}$$

Substituting the above equations and the ideal fluid energy-momentum tensor into the Einstein field equations (2.9), we obtain the Friedmann equations:

$$\boxed{\begin{aligned} \frac{8\pi G}{3}\rho &= H^2 + \frac{Kc^2}{a^2}, & (2.13)\\ -\frac{8\pi G}{c^2} P &= \frac{2\ddot{a}}{a} + H^2 + \frac{Kc^2}{a^2}, & (2.14) \end{aligned}}$$

where $H \equiv \dot{a}/a$ is the Hubble parameter, which quantifies the rate of cosmic expansion.

Subtracting the first Friedmann equation from the second, we derive the acceleration equation:

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3P}{c^2}\right). \tag{2.15}$$

This equation reveals a profound result: the accelerating expansion of the Universe ($\ddot{a} > 0$) requires $\rho + 3P/c^2 < 0$, i.e., a cosmic component with negative pressure.

A fundamental geometric result in GR is the contracted second Bianchi identity, which enforces that the covariant divergence of the Einstein tensor vanishes identically for any metric:

$$\nabla^\mu G_{\mu\nu} = 0. \tag{2.16}$$

Taking the covariant divergence of both sides of the Einstein field equations (2.9), the left-hand side vanishes via the identity above, which yields the covariant energy-momentum conservation law for the total cosmic matter-energy content:

$$\nabla_\mu T^{\mu\nu} = 0. \tag{2.17}$$

For the homogeneous and isotropic FLRW background spacetime, the spatial components ($\nu = 1, 2, 3$) of the covariant conservation law reduce to trivial identities, while the time component ($\nu = 0$) gives the cosmological continuity equation:

$$\boxed{\dot{\rho} + 3H\left(\rho + \frac{P}{c^2}\right) = 0.} \tag{2.18}$$

It is important to note that the continuity equation (2.18) follows directly from the two Friedmann equations (2.13) and (2.14), meaning the three equations are not independent.

2.1.4 Space-Time Dynamics

We hence have two independent equations for three unknowns: the scale factor, the energy density, and the pressure. To solve this system, the most convenient way is to introduce an equation of state (EoS) for the matter content, via the dimensionless equation-of-state parameter defined as:

$$w \equiv \frac{P}{\rho c^2}. \tag{2.19}$$

By differentiating w with respect to cosmic time and combining the continuity equation (2.18), the evolution equation for the EoS parameter is:

$$\dot{w} = -3H(1+w)\left(\frac{c_s^2}{c^2} - w\right), \tag{2.20}$$

where $c_s^2 \equiv \left(\frac{\partial P}{\partial \rho}\right)_s$ is the adiabatic sound speed squared of the fluid. For a barotropic fluid with $P = P(\rho)$, the relation $c_s^2 = wc^2$ holds, such that $\dot{w} = 0$ and the EoS parameter w remains constant over cosmic time.

Substituting the EoS (2.19) into the continuity equation (2.18), we obtain an analytic solution for the evolution of energy density with scale factor for a constant EoS parameter w:

$$\rho(a) = \rho_0 \left(\frac{a_0}{a}\right)^{3(1+w)}, \tag{2.21}$$

where ρ_0 is the component's present-day energy density. From this point onward, we use the subscript 0 to denote quantities evaluated at the present day. This result is extended to the Universe's dominant energy components:

- Non-relativistic matter (cold dark matter + baryons, $w_{\mathrm{m}} = 0$): Thermal pressure is negligible relative to rest mass energy density ($P \approx 0$), giving the scaling:

$$\rho_{\mathrm{m}}(a) = \rho_{\mathrm{m}0} a^{-3}. \tag{2.22}$$

  This arises from the dilution of particle number density ($\propto a^{-3}$) with cosmic expansion, as the rest mass energy per particle remains constant.

- Relativistic radiation (photons + relativistic neutrinos, $w_{\mathrm{r}} = 1/3$): Fixed by statistical mechanics, the EoS gives the scaling:

$$\rho_{\mathrm{r}}(a) = \rho_{\mathrm{r}0} a^{-4}. \tag{2.23}$$

  This comes from volume dilution of particle number ($\propto a^{-3}$) combined with cosmic redshift of particle energy ($E \propto a^{-1}$), yielding the total a^{-4} dependence.

- Dark energy: For a general time-dependent EoS $w_{\mathrm{DE}}(a) = P_{\mathrm{DE}}(a)/(\rho_{\mathrm{DE}}(a)c^2)$, the continuity equation gives:

$$\rho_{\mathrm{DE}}(a) = \rho_{\mathrm{DE}0} \exp\left(-3\int_1^a \frac{1 + w_{\mathrm{DE}}(a')}{a'} \mathrm{d}a'\right). \tag{2.24}$$

The total energy density of the Universe at any scale factor a is the sum of the energy densities of all components:

$$\rho_{\mathrm{tot}}(a) = \rho_{\mathrm{m}}(a) + \rho_{\mathrm{r}}(a) + \rho_{\mathrm{DE}}(a) + \rho_{\mathrm{other}}(a), \tag{2.25}$$

where $\rho_{\mathrm{other}}(a)$ denotes subdominant components (e.g., late-Universe massive neutrinos). The relative contribution of each component evolves significantly with cosmic expansion, driven by their distinct scaling relations. Here, we introduce the dimensionless energy parameters:

$$\Omega_{\mathrm{tot}} = \frac{\rho_{\mathrm{tot}}(a)}{\rho_{\mathrm{crit}}}, \quad \Omega_{\mathrm{K}} = -\frac{Kc^2}{H^2a^2} = -\frac{kc^2}{H^2a^2R_c^2}, \quad R_c = \frac{c}{H_0\sqrt{|\Omega_{\mathrm{K}0}|}}. \tag{2.26}$$

with the critical energy density $\rho_{\mathrm{crit}} \equiv 3H^2/(8\pi G)$ and the density parameter of each component $\Omega_i = \rho_i/\rho_{\mathrm{crit}}$, such that $\Omega_{\mathrm{tot}} = \sum_i \Omega_i$. The first Friedmann equation (2.13) then takes the form of a constraint:

$$\boxed{1 = \sum_i \Omega_i + \Omega_{\mathrm{K}}.} \tag{2.27}$$

2.2 Cosmological Perturbation Theory

While the homogeneous and isotropic FLRW background model successfully describes the mean expansion history of the Universe, it is insufficient to explain the formation of cosmic large-scale structure or the full anisotropy spectrum of the CMB. To address this, we need the evolution of small linear deviations from the background spacetime and energy-momentum content, formalized by cosmological perturbation theory.

In what follows, we make extensive use of conformal time τ, defined by the differential relation $\mathrm{d}\tau = \mathrm{d}t/a(t)$. Throughout this section, we adopt natural units $c = \hbar = 1$ and focus on the spatially flat Universe ($K = 0$) for simplicity. Under this convention, the FLRW metric in conformal time reads:

$$\mathrm{d}s^2 = a(\tau)^2 \left(-\mathrm{d}\tau^2 + \gamma_{ij}\mathrm{d}x^i\mathrm{d}x^j\right). \tag{2.28}$$

The derivative of any quantity X with respect to τ, denoted X', is related to the cosmic time derivative $\dot{X}$ via $\dot{X} = \frac{X'}{a}$. In conformal time, the Friedmann equations take the form:

$$\boxed{\begin{aligned} \mathcal{H}^2 + K &= \frac{8\pi G}{3}\rho a^2, \\ 2\mathcal{H}' + \mathcal{H}^2 + K &= -8\pi G P a^2. \end{aligned}} \begin{aligned} &(2.29) \\ &(2.30) \end{aligned}$$

In terms of conformal time τ, the continuity equation takes the form:

$$\boxed{\rho' + 3\mathcal{H}\left(\rho + P\right) = 0,} \tag{2.31}$$

where $\mathcal{H} \equiv \frac{a'}{a} = aH$ is the conformal Hubble parameter.

All physical quantities $Q(t, \mathbf{x})$ are decomposed into a homogeneous background component $\bar{Q}(t)$ and a small linear perturbation component $\delta Q(t, \mathbf{x})$:

$$Q(t, \mathbf{x}) = \bar{Q}(t) + \delta Q(t, \mathbf{x}), \qquad \left|\frac{\delta Q}{\bar{Q}}\right| \ll 1. \tag{2.32}$$

We also introduce the density contrast $\delta \equiv \delta\rho/\bar{\rho}$ for normalized energy density perturbations.

In Fourier space, we adopt the standard cosmological convention:

$$f(\vec{x}, \tau) = \int \frac{d^3k}{(2\pi)^3} e^{i\vec{k}\cdot\vec{x}} f(\vec{k}, \tau),$$

with $k = |\vec{k}|$ the comoving wavenumber and $\hat{k}^i = k^i/k$ the unit wavevector.

2.2.1 Perturbed Spacetime

A foundational result of linear cosmological perturbation theory is the Scalar-Vector-Tensor (SVT) decomposition, which uniquely decomposes the general perturbed metric into three mutually orthogonal, spin-classified components that fully decouple at linear order in a homogeneous and isotropic background. The SVT-decomposed metric takes the form:

$$\mathrm{d}s^2 = (\bar{g}_{\mu\nu} + \delta g_{\mu\nu})\, \mathrm{d}x^\mu \mathrm{d}x^\nu \tag{2.33}$$

$$= a(\tau)^2 \left[-(1+2\phi)\mathrm{d}\tau^2 + 2\left(\partial_i B + S_i\right) \mathrm{d}\tau \mathrm{d}x^i + \left(\gamma_{ij} + h_{ij}\right) \mathrm{d}x^i \mathrm{d}x^j \right], \tag{2.34}$$

where:

- $h_{ij} = -2\psi\gamma_{ij} + 2\left(\partial_i\partial_j - \frac{1}{3}\gamma_{ij}\nabla^2\right) E + \partial_i F_j + \partial_j F_i + h_{ij}^{\mathrm{TT}}$;
- ϕ, B, ψ, E denote scalar perturbations;
- S_i, F_i represent transverse vector perturbations satisfying $\partial^i S_i = 0$ and $\partial^i F_i = 0$;
- h_{ij}^{TT} is the traceless transverse tensor perturbation obeying $\partial^i h_{ij}^{\mathrm{TT}} = 0$ and $h_{ij}^{\mathrm{TT}}\gamma^{ij} = 0$.

The components of the perturbed metric are thus given by:

$$\boxed{\begin{aligned} \delta g_{00} &= -2a^2(\tau)\phi, && (2.35)\\ \delta g_{0i} &= a^2(\tau)\left(\partial_i B + S_i\right), && (2.36)\\ \delta g_{ij} &= a^2(\tau)\left[-2\psi\gamma_{ij} + 2\left(\partial_i\partial_j - \frac{1}{3}\gamma_{ij}\nabla^2\right) E + \partial_i F_j + \partial_j F_i + h_{ij}^{\mathrm{TT}}\right]. && (2.37) \end{aligned}}$$

For non-vanishing spatial curvature K, partial derivatives are replaced by covariant derivatives compatible with the curved 3-metric γ_{ij}, while the structure of the SVT decomposition remains unchanged.

From the energy-momentum tensor of the perfect fluid in Eq. (2.7), the perturbed energy-momentum tensor of the fluid takes the general form:

$$\delta T_{\mu\nu} = \left(\delta\rho + \delta P\right) \bar{u}_\mu \bar{u}_\nu + \delta P \bar{g}_{\mu\nu} + 2(\bar{\rho} + \bar{P})\bar{u}_{(\mu}\delta u_{\nu)} + \bar{P}\delta g_{\mu\nu} + a^2 \bar{P}\pi_{\mu\nu}, \tag{2.38}$$

where $\pi_{\mu\nu}$ is the traceless anisotropic stress tensor, which quantifies the deviation of the perturbed fluid from a perfect fluid and satisfies $\bar{g}^{\mu\nu}\pi_{\mu\nu} = 0$. The tensor $\pi_{\mu\nu}$ is symmetric and orthogonal to the four-velocity ($\bar{u}^\mu \pi_{\mu\nu} = 0$), so we set $\pi_{00} = \pi_{0i} = 0$ without loss of generality, leaving only spatial components. It decomposes into scalar, vector and tensor parts as:

$$\pi_{ij} = \left(\partial_i\partial_j - \frac{1}{3}\gamma_{ij}\nabla^2\right)\pi + \left(\partial_i\pi_j + \partial_j\pi_i - \frac{2}{3}\gamma_{ij}\gamma^{kl}\partial_k\pi_l\right) + \pi_{ij}^{\mathrm{TT}}, \tag{2.39}$$

where π denotes the scalar anisotropic stress potential, π_i the transverse vector component ($\partial^i\pi_i = 0$), and π_{ij}^{TT} the traceless transverse tensor component.

For a comoving observer, the four-velocity $u^\mu = \bar{u}^\mu + \delta u^\mu$ satisfies $u_\mu u^\mu = -1$, leading to the linear-order perturbation:

$$\delta u^\mu = a^{-1}(-\phi, v^i), \quad \delta u_\mu = a(-\phi, v_i + \partial_i B + S_i), \tag{2.40}$$

where $\delta u^i \equiv v^i/a$, and the spatial velocity perturbation splits into a scalar and a transverse vector part: $v_i = \partial_i v + \bar{v}_i$ with $\partial^i\bar{v}_i = 0$.

Combining Eqs. (2.38), (2.39), and (2.40), the components of the perturbed energy-momentum tensor are:

$$\boxed{\begin{aligned} \delta T_{00} &= \bar{\rho}a^2\left(\delta + 2\phi\right), && (2.41)\\ \delta T_{0i} &= -\bar{\rho}a^2\left[(1+w)\left(\partial_i v + \bar{v}_i\right) + \partial_i B + S_i\right], && (2.42)\\ \delta T_{ij} &= \bar{P}a^2\left(h_{ij} + \frac{\delta P}{\bar{P}}\gamma_{ij} + \pi_{ij}\right). && (2.43) \end{aligned}}$$

where $w \equiv \bar{P}/\bar{\rho}$ is the background equation-of-state parameter.

The Gauge Problem, Gauge-Invariant Quantities, and Gauge Fixing

A central technical challenge in cosmological perturbation theory is the gauge problem: linear perturbations are defined with respect to a chosen coordinate system, and infinitesimal coordinate transformations (gauge transformations) can map the homogeneous background into unphysical, spurious perturbations.

Under an infinitesimal gauge transformation

$$x^\mu \to \tilde{x}^\mu = x^\mu + \xi^\mu(\tau, \mathbf{x}), \quad \xi^\mu \equiv (\alpha, \xi^i), \tag{2.44}$$

where the spatial part of the gauge vector admits the scalar-transverse decomposition

$$\xi^i = \gamma^{ij}\partial_j\beta + L^i, \quad \partial_i L^i = 0, \tag{2.45}$$

the metric transforms according to the general covariance rule

$$\tilde{g}_{\mu\nu}(\tilde{x}) = g_{\rho\sigma}(x)\frac{\partial x^\rho}{\partial\tilde{x}^\mu}\frac{\partial x^\sigma}{\partial\tilde{x}^\nu}. \tag{2.46}$$

At linear order, this yields the gauge transformation rules for the metric perturbation components. Several key properties hold:

- Tensor perturbations h_{ij}^{TT} are fully gauge-invariant: coordinate transformations do not affect traceless-transverse tensors, so gravitational wave modes are physical and coordinate-independent.

- Vector perturbations transform only within the vector sector, with no mixing to scalar or tensor perturbations.

- Scalar perturbations ϕ, B, ψ, E transform non-trivially and exhibit spurious mixing between background quantities and fluctuations:

$$\begin{aligned} \tilde{\phi} &= \phi - \alpha' - \mathcal{H}\alpha, \quad \tilde{B} = B + \alpha - \beta', \\ \tilde{\psi} &= \psi + \mathcal{H}\alpha + \frac{1}{3}\nabla^2\beta, \quad \tilde{E} = E - \beta. \end{aligned} \tag{2.47}$$

Under the same gauge transformation, the linear-order perturbations of the energy density, pressure, and scalar velocity potential of the cosmic fluid obey the following transformation laws:

$$\begin{aligned} \tilde{\delta\rho} &= \delta\rho - \bar{\rho}'\alpha, \\ \tilde{\delta P} &= \delta P - \bar{P}'\alpha, \\ \tilde{v} &= v + \beta'. \end{aligned} \tag{2.48}$$

Like the metric scalar potentials, the perturbed fluid variables $\delta\rho, \delta P, v$ are individually gauge-dependent and do not represent physical observables.

To resolve these ambiguities, one must construct gauge-invariant quantities, which remain invariant under all infinitesimal coordinate transformations and describe only genuine physical fluctuations. Here we present some commonly defined gauge-invariant observables widely used in cosmology.

- **Bardeen Potentials**: Introduced by Bardeen[13], the gauge-invariant potentials Ψ and Φ extend Newtonian gravity to relativistic cosmological perturbations, defined as:

$$\Psi \equiv \phi + \mathcal{H}(B - E') + (B - E')', \quad \Phi \equiv \psi + \frac{1}{3}\nabla^2 E - \mathcal{H}(B - E'), \tag{2.49}$$

  which are strictly invariant under the linear gauge transformations in Eqs. (2.47) and (2.48). In the absence of anisotropic stress, the two potentials are equal ($\Psi = \Phi$), recovering the Newtonian limit of a single gravitational potential.

- **Anisotropic Stress**: The traceless spatial component of the perturbed energy-momentum tensor, which quantifies the deviation from a perfect fluid, is defined as

$$\Pi^i{}_j \equiv \delta T^i_j - \frac{1}{3}\gamma^i{}_j\, \delta T^k_k, \tag{2.50}$$

where $\Pi^i{}_j$ is symmetric, traceless, and gauge-invariant. Its scalar component is described by the scalar anisotropic stress potential Π, via the general curvature-compatible decomposition

$$\Pi^i{}_j = \left(\nabla^i\nabla_j - \frac{1}{3}\gamma^i{}_j\nabla^2\right)\Pi, \tag{2.51}$$

and $\Pi = \bar{P}a^2\pi$ with π the scalar anisotropic stress potential defined in Eq. (2.39).

Under the Fourier transform, spatial covariant derivatives satisfy the standard correspondence

$$\nabla_i \mapsto ik_i, \quad \nabla^i \mapsto ik^i, \quad \nabla^2 = \nabla^i\nabla_i \mapsto -k^2,$$

Substituting this into the coordinate-space decomposition, we obtain:

$$\begin{aligned}
\Pi^i{}_j(\vec{k},\tau) &= \left((ik^i)(ik_j) - \frac{1}{3}\gamma^i{}_j(-k^2)\right)\Pi(\vec{k},\tau) \\
&= \left(-k^ik_j + \frac{1}{3}\gamma^i{}_jk^2\right)\Pi(\vec{k},\tau) \\
&= -k^2\left(\hat{k}^i\hat{k}_j - \frac{1}{3}\gamma^i{}_j\right)\Pi(\vec{k},\tau).
\end{aligned}$$

To extract the scalar potential Π from the tensorial anisotropic stress, we introduce the symmetric, traceless mixed-index projection operator

$$P^i_j = \hat{k}^i\hat{k}_j - \frac{1}{3}\gamma^i{}_j,$$

which is normalized such that $P^i_jP^j_i = 2/3$:

$$\begin{aligned}
P^i_jP^j_i &= \left(\hat{k}^i\hat{k}_j - \frac{1}{3}\gamma^i{}_j\right)\left(\hat{k}^j\hat{k}_i - \frac{1}{3}\gamma^j{}_i\right) \\
&= \hat{k}^i\hat{k}_j\hat{k}^j\hat{k}_i - \frac{1}{3}\hat{k}^i\hat{k}_j\gamma^j{}_i - \frac{1}{3}\gamma^i{}_j\hat{k}^j\hat{k}_i + \frac{1}{9}\gamma^i{}_j\gamma^j{}_i \\
&= (\hat{k}^i\hat{k}_i)(\hat{k}_j\hat{k}^j) - \frac{1}{3}(\hat{k}^i\hat{k}_i) - \frac{1}{3}(\hat{k}_i\hat{k}^i) + \frac{1}{9}\cdot 3 \\
&= 1 - \frac{1}{3} - \frac{1}{3} + \frac{1}{3} \\
&= \frac{2}{3}.
\end{aligned}$$

Contracting the projection operator with both sides of the Fourier-space anisotropic stress tensor, we get

$$\begin{aligned}
P^i_j \Pi^j_i(\vec{k}, \tau) &= P^i_j \left[-k^2 P^j_i \Pi(\vec{k}, \tau)\right] \\
&= -k^2 \Pi(\vec{k}, \tau) \cdot P^i_j P^j_i \\
&= -k^2 \Pi(\vec{k}, \tau) \cdot \frac{2}{3} \\
&= -\frac{2}{3} k^2 \Pi(\vec{k}, \tau).
\end{aligned}$$

Solving for $\Pi(\vec{k}, \tau)$ yields the direct conversion relation:

$$\boxed{\Pi(\vec{k}, \tau) = -\frac{3}{2k^2} P^i_j \Pi^j_i(\vec{k}, \tau) = -\frac{3}{2k^2} \left(\hat{k}^i \hat{k}_j - \frac{1}{3}\gamma^i_j\right) \Pi^j_i(\vec{k}, \tau).} \tag{2.52}$$

- **Comoving Curvature Perturbation**: Defined as the intrinsic spatial curvature perturbation on hypersurfaces comoving with the cosmic fluid, it is a strictly gauge-invariant quantity that remains conserved on super-horizon scales for adiabatic perturbations. For our SVT decomposition, it is given by

$$\zeta \equiv \psi + \frac{1}{3}\nabla^2 E + \mathcal{H}\frac{\delta\rho}{\bar{\rho}'}. \tag{2.53}$$

  As a key bridge between early-Universe inflationary physics and late-time cosmological observables, ζ is the primary quantity used to characterize primordial density fluctuations.

An alternative solution to the gauge problem is gauge fixing—choosing a specific coordinate system to eliminate spurious gauge degrees of freedom and simplify perturbative calculations, with different gauges optimized for distinct cosmological scenarios. The most widely used gauges are listed below, with the fully gauge-invariant transverse-traceless tensor perturbations h^{TT}_{ij} remaining unconstrained in all cases.

- **Newtonian Gauge**: The most intuitive gauge for mapping relativistic perturbations to Newtonian gravity, fixed by $B = 0$, $E = 0$ and $S_i = 0$, $F_i = 0$. It yields a strictly diagonal metric, with scalar perturbations ϕ, ψ exactly matching the gauge-invariant Bardeen potentials Ψ, Φ. This gauge is widely used for late-Universe large-scale structure formation studies, CMB anisotropy calculations.

- **Spatially Flat Gauge**: Fixed by $\psi = 0$, $E = 0$ and $F_i = 0$. It eliminates scalar spatial curvature perturbations to isolate the time-time metric perturbation ϕ, and is particularly useful for early-Universe perturbation studies and inflationary fluctuation generation.

- **Synchronous Gauge**: Fixed by $\phi = 0$, $B = 0$ and $S_i = 0$. First introduced by Lifshitz [14], it aligns coordinate time with the proper time of comoving observers, making it the standard choice for numerical CMB Boltzmann codes. A notable limitation is that it retains residual time-independent spatial gauge freedom, which requires careful handling in numerical calculations.

- **Uniform Density Gauge**: Fixed by $\delta\rho = 0$, $B = 0$ and $S_i = 0$. In this gauge, the spatial metric perturbation directly maps to the comoving curvature perturbation ζ, making it ideal for analyzing inflationary primordial fluctuations and super-horizon mode evolution.

- **Comoving Gauge**: Fixed by $v + B = 0$ and $\bar{v}_i + S_i = 0$. It ties the coordinate frame directly to the comoving motion of the cosmic fluid, with $\psi + \frac{1}{3}\nabla^2 E = \zeta$ for adiabatic perturbations, making it the canonical gauge for inflationary cosmology [15].

2.2.2 Perturbed Einstein and Continuity Equations

SVT decomposition of the perturbed metric and energy-momentum tensor, followed by substitution into Einstein's field equations and covariant energy-momentum conservation, yields the linearized evolution equations for cosmological perturbations. In a homogeneous, isotropic background, these SVT modes fully decouple at first order, permitting independent analysis of each sector. We focus first on scalar perturbations, the component most directly tied to observational cosmology.

The full Einstein field equations read $G_{\mu\nu} = 8\pi G T_{\mu\nu}$, with both tensors split at linear order into background and perturbed parts:

$$G_{\mu\nu} = \bar{G}_{\mu\nu} + \delta G_{\mu\nu}, \quad T_{\mu\nu} = \bar{T}_{\mu\nu} + \delta T_{\mu\nu}. \tag{2.54}$$

Since the unperturbed background satisfies $\bar{G}_{\mu\nu} = 8\pi G \bar{T}_{\mu\nu}$, subtracting the background relation from the full equations isolates the first-order linearized Einstein field equations:

$$\boxed{\delta G_{\mu\nu} = 8\pi G \delta T_{\mu\nu}.} \tag{2.55}$$

Scalar perturbations are the dominant driver of structure formation and CMB anisotropies. To eliminate gauge ambiguity, we present the equations in terms of the gauge-invariant Bardeen potentials Ψ and Φ (Eq. (2.49)). Substituting the scalar components of $\delta g_{\mu\nu}$ (Eq. (2.37)) and $\delta T_{\mu\nu}$ into Eq. (2.55) yields four independent equations:

1. **Time-Time Component (δG_{00}): Poisson Equation**
   relating spatial curvature perturbations to energy density fluctuations:

$$\boxed{\nabla^2\Phi - 3\mathcal{H}\left(\Phi' + \mathcal{H}\Psi\right) = 4\pi G a^2 \delta\rho.} \tag{2.56}$$

On sub-horizon scales ($k \gg \mathcal{H}$, where k is the comoving wavenumber and $\nabla^2 \sim -k^2$ in Fourier space), the Laplacian term is dominant, reducing the equation to a direct relation in Newtonian limit.

2. **Time-Space Component (δG_{0i}):**
This equation links the time evolution of the gravitational potentials to the bulk fluid velocity:

$$\boxed{\partial_i \left(\Phi' + \mathcal{H}\Psi\right) = -4\pi G a^2 (\bar{\rho} + \bar{P}) \partial_i v.} \tag{2.57}$$

Dropping the common gradient operator ∂_i gives the scalar form:

$$\Phi' + \mathcal{H}\Psi = -4\pi G a^2 (\bar{\rho} + \bar{P}) v. \tag{2.58}$$

3. **Trace of Space-Space Component (δG_{ii}):**
This equation governs the time evolution of the Bardeen potentials, sourced by pressure perturbations:

$$\boxed{\Phi'' + \mathcal{H}\left(2\Phi' + \Psi'\right) + \left(2\mathcal{H}' + \mathcal{H}^2\right)\Psi + \frac{1}{3}\nabla^2\left(\Psi - \Phi\right) = 4\pi G a^2 \delta P.} \tag{2.59}$$

4. **Traceless Space-Space Component: Anisotropic Stress Constraint**
This equation encodes the relation between the two Bardeen potentials and the scalar anisotropic stress potential Π, which is the only source of the difference between Ψ and Φ:

$$\boxed{\left(\partial_i\partial_j - \frac{1}{3}\gamma_{ij}\nabla^2\right)(\Phi - \Psi) = 8\pi G \left(\partial_i\partial_j - \frac{1}{3}\gamma_{ij}\nabla^2\right) a^2 \Pi.} \tag{2.60}$$

Removing the common differential operator recovers the gauge-invariant relation:

$$\Phi - \Psi = 8\pi G a^2 \Pi. \tag{2.61}$$

For an ideal fluid with $\Pi = 0$, $\Psi = \Phi$, reducing to a single gravitational potential as in Newtonian gravity.

Linearized Energy-Momentum Conservation

As in the background FLRW cosmology, the contracted second Bianchi identity $\nabla^\mu G_{\mu\nu} = 0$ ensures that the linearized Einstein equations automatically enforce the covariant conservation of the perturbed energy-momentum tensor:

$$\boxed{\nabla^\mu \delta T_{\mu\nu} = 0.} \tag{2.62}$$

For scalar perturbations, this equation splits into two independent components:

1. **Time Component ($\nu = 0$): Perturbed Continuity Equation**
   This equation describes the evolution of energy density fluctuations, generalizing the background continuity equation to linear order:

$$\boxed{\delta\rho' + 3\mathcal{H}\left(\delta\rho + \delta P\right) + \left(\bar{\rho} + \bar{P}\right)\left(\partial_i v^i - \psi'\right) = 0.} \tag{2.63}$$

2. **Space Component ($\nu = i$): Euler Equation**
   This equation governs the evolution of fluid velocity perturbations, generalizing the Euler equation to linear order, including the contribution from anisotropic stress:

$$\boxed{\left[(\bar{\rho} + \bar{P})v_i\right]' + 4\mathcal{H}(\bar{\rho} + \bar{P})v_i + (\bar{\rho} + \bar{P})\partial_i \Psi + \partial_i \delta P + \partial^j \Pi^i_j = 0.} \tag{2.64}$$

Together, the linearized Einstein equations and the perturbed continuity/Euler equations form a complete system for evolving perturbations from the early Universe to the present day, providing the theoretical foundation for studies of structure formation and CMB anisotropies.

2.2.3 Adiabatic Perturbations and Entropy Perturbations

The origin of primordial perturbations lies in the physics of the very early Universe. While various mechanisms (e.g., inflation, cosmic strings, or bouncing cosmologies) have been proposed to generate these fluctuations, their classification and general properties can be analyzed in a model-independent manner.

In this section, we provide a brief review of adiabatic and entropy primordial perturbations.

Adiabatic Perturbations

Adiabatic perturbations correspond to vanishing relative entropy fluctuations between different cosmic fluid components. At linear order, the specific entropy of each species exhibits identical spatial perturbations, such that the relative entropy perturbation between any two components satisfies

$$\boxed{0 = \delta s_{ij} \equiv s_i - s_j = \frac{\delta n_i}{\bar{n}_i} - \frac{\delta n_j}{\bar{n}_j} = \frac{\delta_i}{1 + w_i} - \frac{\delta_j}{1 + w_j} \quad \forall\, i, j,} \tag{2.65}$$

where $\delta_i = \delta\rho_i/\bar{\rho}_i$ denotes the density contrast of species i. Physically, they describe a spatially varying time delay in cosmic expansion, with the density perturbation ratio $\delta_i/(1 + w_i)$ uniform across all species.

Expanding the time derivative of the comoving curvature perturbation in Eq. (2.53) and substituting the linearized cosmological perturbation equations yields the full evolution equation:

$$\zeta' = -\frac{\mathcal{H}}{\bar{\rho}+\bar{P}}\nabla^2\left(\frac{\bar{\rho}+\bar{P}}{\mathcal{H}^2}v\right). \tag{2.66}$$

On super-horizon scales ($k \ll \mathcal{H}$), the spatial gradient term vanishes identically, leading to the key conservation law:

$$\boxed{\zeta' = 0 \quad \text{for} \quad k \ll \mathcal{H},} \tag{2.67}$$

This conservation law establishes a direct connection between early-Universe primordial fluctuations and late-time cosmological observables, and holds for adiabatic perturbations in both single-component cosmic fluids and multi-component fluids with strictly adiabatic initial conditions.

For vanishing anisotropic stress, ζ is related to the Bardeen potential Φ through

$$\boxed{\zeta = \frac{5+3w}{3(1+w)}\Phi \quad (\text{super-horizon, adiabatic, } \Pi = 0).} \tag{2.68}$$

Entropy (Isocurvature) Perturbations

Isocurvature perturbations correspond to fluctuations in the relative abundance of cosmic fluid components, with zero total energy density perturbation (and thus vanishing initial spatial curvature) at the time of generation:

$$\boxed{\delta\rho_{\text{tot}} = \sum_i \delta\rho_i = 0.} \tag{2.69}$$

They are defined by non-vanishing relative entropy fluctuations between species:

$$\boxed{\delta s_{ij} \equiv s_i - s_j = \frac{\delta n_i}{\bar{n}_i} - \frac{\delta n_j}{\bar{n}_j} = \frac{\delta_i}{1+w_i} - \frac{\delta_j}{1+w_j},} \tag{2.70}$$

Typical isocurvature modes include relative fluctuations between cold dark matter and baryons, or between photons and neutrinos. In contrast to adiabatic perturbations, isocurvature modes are not conserved on super-horizon scales and can source curvature perturbations at horizon crossing as the Universe expands. Current CMB data place stringent upper limits on isocurvature amplitudes, confirming a strongly adiabatically dominated primordial spectrum [16–18].

Chapter 3

COSMOLOGICAL OBSERVABLES

Building on the theoretical framework in the previous chapter, we now turn to the key cosmological observational probes that connect theoretical predictions to cosmological measurements. In this chapter, we present the interpretative framework for the core observations employed in this work: cosmic microwave background (CMB) anisotropies, redshift-space distortions (RSD), baryon acoustic oscillations (BAO), and Type Ia supernovae (SNe Ia).

3.1 Cosmic Microwave Background Anisotropies

The CMB originated when neutral hydrogen formed during the epoch of recombination, freeing primordial photons from the dense baryon-electron plasma to travel unimpeded across the expanding Universe to the present day. This radiation encodes the fundamental physics of the early Universe and serves as a snapshot of primordial conditions at the last scattering surface (LSS). The CMB exhibits a near-perfect blackbody spectrum and a uniform 2.73 K all-sky temperature, its extreme homogeneity and isotropy confirming the cosmological principle on the largest cosmic scales. Its tiny temperature fluctuations at the level of $\Delta T/T \sim 10^{-5}$ can be explained within various competing early-Universe paradigms. The most widely accepted framework for their origin combines inflation with quantum field theory, where these primordial density ripples seed all observed large-scale structure in the Universe.

The dominant linear contributions to the CMB power spectrum arise from two core physical mechanisms: the Sachs–Wolfe (SW) gravitational redshift effect, which includes the integrated Sachs–Wolfe (ISW) effect, and Doppler velocity effects. At linear order, the full temperature fluctuation can be rigorously decomposed using the line-of-sight integral formalism as [19]:

$$\frac{\delta T}{\bar{T}}(\hat{\mathbf{n}}) = \underset{\substack{\uparrow \\ \text{SW}}}{\left(\frac{1}{4}\delta_\gamma + \Psi\right)_*} - \underset{\substack{\uparrow \\ \text{Doppler}}}{(\hat{\mathbf{n}}\cdot\mathbf{v}_b)_*} + \underset{\substack{\uparrow \\ \text{ISW}}}{\int_{\tau_*}^{\tau_0} \mathrm{d}\tau\,(\Psi' + \Phi')}, \tag{3.1}$$

where $\hat{\mathbf{n}}$ is the unit vector along the observer's line of sight. The subscript $_*$ denotes quantities evaluated at the LSS, $\mathbf{v}_b$ is the bulk peculiar velocity of the baryon-photon plasma. The first two terms on the right-hand side of Eq. (3.1) correspond to the primary anisotropies imprinted at the LSS, while the final integral term describes secondary anisotropies accumulated along the photon's path from the LSS to the present epoch.

3.1.1 Sachs-Wolfe Effect

The primary Sachs-Wolfe (SW) effect dominates the primary CMB temperature anisotropies on the largest angular scales $\ell \lesssim 20$, arising from gravitational redshift induced by primordial gravitational potentials at recombination. It corresponds to the first term in Eq. (3.1):

$$\left(\frac{1}{4}\delta_\gamma + \Psi\right)_*. \tag{3.2}$$

These large-scale modes satisfy $k\tau_* \ll 1$, indicating they remained outside the cosmic horizon at recombination and retained their primordial form until last scattering.

According to adiabatic initial conditions (2.65), for photons, baryons, and cold dark matter (with no isocurvature contributions):

$$\frac{\delta_\gamma}{4} = \frac{\delta_b}{3} = \frac{\delta_c}{3}. \tag{3.3}$$

Combining the fact that the comoving curvature perturbation ζ is rigorously conserved on superhorizon scales for adiabatic fluctuations (2.67) and assuming a matter-dominated Universe at recombination, the linearized Einstein equations relate the photon perturbation to the gravitational potentials for superhorizon modes. Substituting this relation into Eq. (3.2) yields the general SW formula, which does not require the assumption of vanishing anisotropic stress:

$$\left.\frac{\delta T}{\bar{T}}\right|_{\text{SW}} = \frac{2\Psi - \Phi}{3}. \tag{3.4}$$

This reduces to the canonical result $\Phi_*/3$ for negligible anisotropic stress. The SW effect directly encodes the primordial gravitational potential at the LSS, making it a critical probe of the initial conditions set by the physics of the early Universe.

3.1.2 Doppler Effect

Doppler shifts induced by bulk peculiar motions in the baryon–photon plasma at the last scattering surface shape the characteristic acoustic peaks in the CMB power spectrum at intermediate angular scales $\ell \sim 100$–1000. This effect is formally described by the second term in Eq. (3.1):

$$-\left(\hat{\mathbf{n}} \cdot \mathbf{v}_b\right)_* . \tag{3.5}$$

Prior to recombination, frequent Thomson scattering enforces a tight-coupling regime in which photons and baryons behave as a single coherent fluid. In this limit, baryons and photons share identical peculiar velocities, and acoustic oscillations are driven by the competition between gravitational infall into primordial potential wells and the restoring pressure exerted by the relativistic photon fluid.

The dynamical evolution of velocity perturbations under the tight-coupling approximation is governed by the linearized Euler equation (2.64):

$$\mathbf{v}' + \mathcal{H}\mathbf{v} = -\nabla\Psi. \tag{3.6}$$

For plane-wave harmonic perturbations, a fundamental phase difference of $\pi/2$ exists between velocity and photon density fluctuations, such that the maximum of one quantity coincides with the minimum of the other.

This intrinsic phase shift directly sculpts the acoustic peak structure of the CMB power spectrum. Under adiabatic initial conditions, odd-numbered acoustic peaks correspond to density maxima at recombination and exhibit suppressed Doppler contributions, while even-numbered peaks align with velocity maxima and feature suppressed density contributions. The relative amplitudes of these peaks provide a model-independent probe of the cosmic baryon density, serving as a key consistency test for the standard ΛCDM model. On small scales $\ell \gtrsim 1000$, Doppler anisotropies are exponentially suppressed by Silk damping, during which photon diffusion erases small-scale inhomogeneities within the baryon-photon fluid.

3.1.3 Integrated Sachs-Wolfe Effect

The integrated Sachs-Wolfe (ISW) effect describes secondary temperature anisotropies accumulated as CMB photons propagate through time-evolving gravitational potentials between the last scattering surface and the observer, corresponding to the integral term in Eq. (3.1),[20–23]:

$$\int_{\tau_*}^{\tau_0} \mathrm{d}\tau \left(\Psi' + \Phi'\right). \tag{3.7}$$

This expression makes explicit that the ISW effect is non-zero if either gravitational potential evolves with time ($\Psi' \neq 0$ or $\Phi' \neq 0$). Even in a perfectly matter-dominated universe, a time-evolving anisotropic stress will generate a non-zero ISW signal.

In a perfectly matter-dominated, spatially flat universe with vanishing anisotropic stress and the ISW effect vanishes identically. However, the Universe deviates from pure matter domination at early and late times, giving rise to two distinct ISW contributions, both modified by the presence of anisotropic stress:

1. **Early-time ISW**: This contribution arises during the radiation-dominated epoch and the radiation-to-matter domination transition shortly after recombination. In the radiation-dominated era, free-streaming neutrinos produce a non-vanishing, time-evolving anisotropic stress, modifying the evolution of both Ψ and Φ and enhancing the early ISW signal. This contributes to the CMB power spectrum at intermediate angular scales and serves as a direct probe of the effective number of relativistic species $N_{\rm eff}$[24–26] and early dark energy model[27, 28].

2. **Late-time ISW**: This contribution arises in the recent epoch of cosmic acceleration driven by dark energy. As the Universe transitions from matter to dark energy domination, the accelerated expansion induces decay of both gravitational potentials. This effect dominates on large angular scales ($\ell \lesssim 100$), where the line-of-sight integral accumulates the strongest signal from evolving potentials [29–31].

The late-time ISW effect is a unique probe of dark energy and the late-time expansion history of the Universe. Unlike the primary SW effect, which is directly sourced by primordial perturbations, the ISW effect is correlated with the large-scale structure along the photon line of sight. This cross-correlation has been robustly detected via joint analysis of CMB temperature maps and galaxy surveys[30, 31], providing independent confirmation of cosmic acceleration and tight constraints on dark energy properties.

3.1.4 CMB Lensing

Gravitational lensing of the CMB is the dominant secondary anisotropy at small-to-intermediate angular scales. It arises from the deflection of CMB photons by the gravitational potential of large-scale structure along the line of sight. Unlike primary anisotropies imprinted at recombination, lensing only remaps the primordial CMB temperature and polarization fields without generating new intrinsic fluctuations. It is sensitive to the Weyl potential $(\Psi + \Phi)/2$, providing a unique probe of cosmic structure growth [1, 32, 33].

Formally, lensing remaps the intrinsic unlensed sky direction $\hat{\mathbf{n}}'$ to the observed direction $\hat{\mathbf{n}}$ via the deflection angle $\boldsymbol{\alpha}(\hat{\mathbf{n}}) = \nabla_{\hat{\mathbf{n}}}\phi(\hat{\mathbf{n}})$, where $\phi(\hat{\mathbf{n}})$ is the dimensionless lensing potential, and $\nabla_{\hat{\mathbf{n}}}$ denotes the 2D angular gradient operator on the celestial sphere. The observed temperature fluctuation reads

$$\frac{\delta T^{\mathrm{obs}}}{\bar{T}}(\hat{\mathbf{n}}) = \frac{\delta T^{\mathrm{unlensed}}}{\bar{T}}\left(\hat{\mathbf{n}} - \nabla_{\hat{\mathbf{n}}}\phi(\hat{\mathbf{n}})\right), \tag{3.8}$$

with the same remapping applied to E-mode and B-mode polarization. From the photon geodesic equation in Newtonian gauge, the lensing potential is the line-of-sight integral of the Weyl potential in a flat FLRW universe[34]:

$$\phi(\hat{\mathbf{n}}) = -\frac{1}{2}\int_{\tau_*}^{\tau_0} \mathrm{d}\tau\, \frac{\tau(\tau_0 - \tau)}{\tau_0}\, (\Psi + \Phi)\,, \tag{3.9}$$

where τ_* and τ_0 are the conformal times at last scattering and today, respectively. The geometric prefactor corresponds to the lensing efficiency kernel, describing the relative angular diameter distances between the observer, the lensing structure, and the CMB last scattering surface.

Key observational signatures include the smoothing of acoustic peaks, non-Gaussian correlations that enable lensing reconstruction, E-mode to B-mode polarization conversion, and a lensing power spectrum that tightly constrains structure growth. Lensing-induced B-modes form the dominant foreground for inflationary gravitational-wave searches [35, 36]. **Delensing** mitigates this contamination by reconstructing the lensing potential and undoing the photon deflection. Current Planck and ground-based analyses achieve 30–50% reduction in lensing B-mode power [33], while next-generation experiments (e.g., CMB-S4) target 90% mitigation to probe inflationary physics [37].

3.2 Redshift Space Distortions

While the CMB encodes primordial perturbations and early-Universe physics at recombination, redshift-space distortions (RSD) provide a unique dynamical probe of the late-time growth of cosmic structure. By tracing the evolution of density perturbations across cosmic time, RSD serve as an indispensable tool for constraining the Universe's structure growth history and testing GR [38, 39].

3.2.1 Growth Rate of Cosmic Structure

At linear order in perturbation theory, the anisotropic redshift-space power spectrum $P_{\mathrm{rs}}(k,\mu)$ is described by the Kaiser formula [38] (valid for large scales $k \lesssim 0.2\,h\,\mathrm{Mpc}^{-1}$), which relates it to the isotropic real-space matter power spectrum $P_{\mathrm{m}}(k)$:

$$P_{\mathrm{rs}}(k,\mu) = P_{\mathrm{m}}(k)\left(b + f(z)\,\mu^2\right)^2, \tag{3.10}$$

where $\mu = \hat{\mathbf{k}} \cdot \hat{\mathbf{n}}$ is the cosine of the angle between the wavevector $\mathbf{k}$ and the line of sight, b is the linear galaxy bias, and $f(z)$ is the linear growth rate of structure. The growth rate is defined as the logarithmic derivative of the linear growth factor $D(a)$ with respect to the scale factor a:

$$f(z) = \frac{d \ln D(a)}{d \ln a}, \tag{3.11}$$

where $D(a)$ quantifies the amplitude of linear density perturbations via $\delta(\mathbf{x}, a) = D(a)\delta(\mathbf{x}, 1)$, normalized such that $D(a = 1) = 1$ at the present day.

The combination $f\sigma_8(z)$ can be directly measured from redshift-space distortions, which breaks the strong degeneracy between the growth rate f and galaxy bias b, yielding more robust and tighter constraints on the growth history of cosmic structure and theories of gravity.

3.2.2 Effective Gravitational Coupling and Gravitational Slip

In the absence of anisotropic stress, GR predicts a tight relation between the growth rate $f(z)$ and the cosmic matter density $\Omega_{\mathrm{m}}(z)$, alongside the no-slip condition $\Psi = \Phi$. To quantify departures from GR, we introduce an effective Newton constant $G_{\mathrm{eff}}(a)$, such that the Poisson equation for the gravitational potential Φ on sub-horizon scales becomes

$$k^2\Phi = -4\pi G_{\mathrm{eff}}(a)\, a^2 \delta\rho_m, \tag{3.12}$$

where $G_{\mathrm{eff}} = G$ recovers standard general relativity. A complementary parameter $\Sigma(a)$ governs the Weyl potential $(\Psi + \Phi)/2$, which determines the light deflection probed by gravitational lensing:

$$k^2(\Psi + \Phi) = -8\pi G_{\mathrm{eff}}(a)\, a^2 \delta\rho_m\, \Sigma(a). \tag{3.13}$$

In GR, $\Sigma = 1$ and $\Psi = \Phi$ hold exactly. In modified gravity scenarios, G_{eff} and Σ become redshift-dependent functions, and gravitational slip $\Psi \neq \Phi$ arises naturally. The RSD observable $f\sigma_8(z)$ constrains the combination of $G_{\mathrm{eff}}(a)$ and the gravitational slip Ψ/Φ, since the structure growth rate depends on both the strength of gravitational clustering and the relation between the two Bardeen potentials:

$$f(z) \simeq \Omega_{\mathrm{m}}(z)^{\gamma(z)}, \quad \gamma(z) = \gamma_{\mathrm{GR}} + \Delta\gamma(G_{\mathrm{eff}}, \tau), \tag{3.14}$$

where $\gamma_{\mathrm{GR}} \approx 0.55$ is the GR prediction for the growth index in ΛCDM [40]. By measuring the redshift evolution of $f\sigma_8(z)$ from RSD multipoles, we infer constraints on modified gravity parameters and test consistency with GR.

3.3 Baryon Acoustic Oscillations

In the pre-recombination Universe ($z \gtrsim 1100$), photons and baryons are tightly coupled by Thomson scattering, forming a single baryon-photon fluid. Primordial density perturbations drive compressional and rarefaction acoustic waves in this fluid, propagating at the sound speed c_s until the drag epoch ($z_{\rm d}$).

At the recombination epoch $z_* \approx 1090$, electrons and protons combine to form neutral hydrogen, causing the Thomson scattering rate to drop sharply and photons to decouple from the baryon fluid. Residual coupling between baryons and the remaining free electrons persists until the drag epoch $z_{\rm d} \approx 1060$, when the pressure support from photons is completely lost, abruptly freezing the acoustic waves. This freeze-out imprints a characteristic comoving scale—the sound horizon $r_{\rm d}$ at the drag epoch—as a preferred separation in the clustering of baryonic tracers (galaxies, quasars, Lyα forest absorbers) at late times. Baryon acoustic oscillations (BAO) preserve the frozen signature of acoustic waves propagating in the primordial photon-baryon plasma before the end of the drag epoch, providing a purely geometrical observable anchored to the microphysics of the early Universe [41–44].

The comoving sound horizon at the drag epoch $r_{\rm d}$ is computed by:

$$r_{\rm d} = \int_0^{\tau_{\rm d}} c_s(\tau)\, d\tau, \tag{3.15}$$

where the sound speed of the photon-baryon fluid is

$$c_s(\tau) = \frac{1}{\sqrt{3\left(1 + \frac{3\rho_{\rm b}}{4\rho_\gamma}\right)}} = \frac{1}{\sqrt{3\left(1 + \frac{3\Omega_{\rm b}h^2}{4\Omega_\gamma h^2}a\right)}}. \tag{3.16}$$

The drag epoch redshift $z_{\rm d}$ is determined by the balance between Thomson scattering drag and cosmic expansion:

$$\int_{z_{\rm d}}^{\infty} \frac{\sigma_T n_e(z) a(z)}{H(z)} dz = 1, \tag{3.17}$$

where σ_T is the Thomson cross section, $n_e(z)$ the free electron density.

3.4 Type Ia Supernovae

Type Ia supernovae are calibrated standard candles that enable precise measurements of the luminosity distance $D_{\rm L}(z)$ as a function of redshift. They were the first probe to provide direct evidence for the accelerated expansion of the Universe, leading to the discovery of dark energy [45, 46].

The observed distance modulus is defined as

$$\mu_{\mathrm{obs}}(z) = m_{\mathrm{B}}(z) - M_{\mathrm{B}}, \tag{3.18}$$

where $m_{\mathrm{B}}(z)$ denotes the apparent B-band magnitude, and M_{B} is the absolute B-band magnitude of a fiducial SNe Ia. The theoretical distance modulus $\mu_{\mathrm{model}}(z)$ is computed by

$$\mu_{\mathrm{model}}(z) = 5\log_{10}\left(\frac{D_{\mathrm{L}}(z)}{\mathrm{Mpc}}\right) + 25, \tag{3.19}$$

where the luminosity distance is given by

$$D_{\mathrm{L}}(z) = \frac{(1+z)}{H_0}\,\frac{S_{\mathrm{k}}\left(\sqrt{|\Omega_{\mathrm{k}}|}\,D_{\mathrm{C}}(z)\right)}{\sqrt{|\Omega_{\mathrm{k}}|}}, \tag{3.20}$$

in which $D_{\mathrm{C}}(z)$ is the comoving distance to redshift z:

$$D_{\mathrm{C}}(z) = \int_0^z \frac{H_0\mathrm{d}\tilde{z}}{H(\tilde{z})}. \tag{3.21}$$

The curvature function $S_{\mathrm{k}}(r)$ is defined as

$$S_{\mathrm{k}}(r) = \begin{cases} \sin r & (k=1), \\ \sinh r & (k=-1), \\ r & (k=0). \end{cases}$$

SNe Ia data provide a direct measurement of the expansion history $H(z)$ and the distance–redshift relation. Together, BAO and SNe Ia constitute excellent probes for constraining the cosmic background evolution, especially the present-day expansion rate H_0, making them powerful tools for testing dark energy models and verifying the consistency of the standard cosmological paradigm.

Exercise 1. Plot the luminosity distance $D_{\mathrm{L}}(z)$ as a function of redshift using the Pantheon+ data catalog shown in [47].

Example 3.1. An example is available at `https://github.com/XueZHOU-cosmology/Tutorial`.

Chapter 4

CAMB: Code for Anisotropies in the Microwave Background

CAMB is one of the most widely used numerical codes for calculating linear cosmological perturbations and their observational signatures. The official website `https://camb.info` hosts the latest source code, documentation, and reference materials.

In this chapter, we present the core architecture, equations, and practical modification guidelines for CAMB, aiming to provide a reference for implementing custom cosmological models and producing working modified versions of the code.

4.1 CAMB Core Structure and Files

CAMB consists of two main layers: a Fortran backend for numerical computations, and a Python frontend for parameter management, result extraction, and visualization. Its key file structure is outlined below:

- `docs/`: Official Jupyter notebooks with examples on equation derivations, code usage, result extraction, and visualization.
- `camb/`: CAMB's official Python frontend, offering interfaces for parameter setup, calculation execution, and result post-processing. See `docs/` examples for detailed usage.
- `fortran/`: Core Fortran backend implementing the Einstein-Boltzmann system, including background cosmic evolution, perturbation dynamics for various components (baryons, CDM, photons, neutrinos, dark energy) and types (scalar, vector, tensor), Boltzmann hierarchy solvers, and multiple dark energy models.
- `forutils/`: General-purpose Fortran utility library providing numerical functions for the core. A shared dependency for all computational subroutines to ensure efficiency and reusability.

- `inifiles/`: Predefined `.ini` parameter templates. Define cosmological parameters and calculation settings witshout the Python interface; used for batch jobs and low-level testing on HPC clusters.

In what follows, we present the core equations used in the calculations of background evolution and perturbation equations. For the full derivation of the equations and conventions used in CAMB, we refer the reader to the official interactive notebook https://camb.readthedocs.io/en/latest/ScalEqs.html and the official CAMB note https://cosmologist.info/notes/CAMB.pdf.

4.2 CAMB Fortran Background Evolution

The background evolution module computes the homogeneous expansion history of the universe as a function of the scale factor a. All quantities are defined in conformal time with units of megaparsecs.

`dtauda` As the first function defined in `equations.f90`, `dtauda(a)` returns the derivative of conformal time with respect to the scale factor: $\mathrm{d}\tau/\mathrm{d}a = 1/a'$. It is calculated from the Friedmann equation and is used in `results.f90` to compute the conformal Hubble parameter $\mathcal{H} = a'/a$, the conformal age of the universe, cosmological distances, and the background evolution; by construction, it is guaranteed to remain finite everywhere. We now examine this function in detail.

Note. The commandline 20 gives

$$\frac{1}{a'} \equiv \texttt{dtauda} = \sqrt{\frac{3}{\texttt{grhoa2}}} \quad \Rightarrow \quad \mathcal{H}^2 = \frac{8\pi G}{3}\bar{\rho}_{\mathrm{tot}} a^2, \tag{4.1}$$

which is exactly the Friedmann equation shown in (2.29), where $\bar{\rho}_{\mathrm{tot}}$ here includes the contribution from curvature.

The effective total energy term $\texttt{grhoa2} \equiv 8\pi G\bar{\rho}_{\mathrm{tot}} a^4$ is obtained by

$$\texttt{grhoa2} = \texttt{grho\_no\_de(a)} + \texttt{grhov\_t}\ a^2, \tag{4.2}$$

where `grho_no_de(a)` denotes the sum of contributions from non-dark energy components (matter, radiation, and curvature). The term `grhov_t` originates from the dark energy `.f90` source files.

`grho_no_de(a)` Defined in `results.f90` (line 1215), this function returns the contribution of non-dark energy components (dubbed “node”) to the background density, encompassing matter, radiation, and curvature:

$$\begin{aligned}\texttt{grho\_no\_de(a)} &= \texttt{grhocrit} \cdot a^4 \cdot \Omega_{\mathrm{M}}(a)\, \frac{H^2}{H_0^2}, \\ &= \texttt{grhocrit} \cdot \left[\Omega_{\mathrm{k0}} a^2 + (\Omega_{\mathrm{b0}} + \Omega_{\mathrm{c0}})\, a + \Omega_{\mathrm{g0}} + \Omega_{\mathrm{r0}}\right] + \texttt{grhoa2}_{\nu},\end{aligned}$$

where $\Omega_{\mathrm{M}}(a)$ denotes the total density parameter excluding dark energy, and `grhocrit` is the present-day critical density defined in CAMB's internal units (see `results.f90`, line 439):

$$\texttt{grhocrit} \equiv \frac{(8\pi G)\; a_0^2\; \rho_{\mathrm{crit0}}}{c^2} = \frac{(8\pi G)\; 3H_0^2}{(c/1000)^2} \quad \text{(In CAMB's units, } \kappa = 1, \text{and } H_0 \text{ in km/s/Mpc).}$$

The term $\texttt{grhoa2}_{\nu}$ accounts for the contribution from massive neutrinos. Since the calculation of massive neutrinos is not considered in this work, interested readers are referred to the corresponding sections for details.

`BackgroundDensityAndPressure` Defined in the dark energy `.f90` files (depending on the chosen dark energy model), this subroutine computes the effective background dark energy density `grhov_t` and (optionally) the equation of state $w_{\mathrm{de}}(a)$ at a given scale factor a. Its input is the present-day dark energy density `grhov`, defined as follows (see `results.f90`, line 460):

$$\texttt{grhov} = \texttt{grhocrit} \cdot \texttt{Omega\_de}, \text{ with } \texttt{Omega\_de} \equiv (1 - \Omega_{\mathrm{M0}}).$$

The subroutine returns the time evolution of the effective dark energy quantities as:

$$\begin{aligned}\texttt{grhov\_t} &= \texttt{grhocrit} \cdot a^2 \cdot \Omega_{\mathrm{de}}(a)\, \frac{H^2}{H_0^2}, \\ \text{(optional output)} &\quad w_{\mathrm{de}}(a).\end{aligned}$$

Specifically, the official CAMB `fortran` folder provides four internal example dark energy files:

- `DarkEnergyInterface.f90` (Default): Implements the Chevallier-Polarski-Linder (CPL) parametrization of dark energy, featuring a time-varying equation of state $w(a) = w_0 + w_a(1 - a)$.
- `DarkEnergyFluid.f90`: Implements a fluid model of dark energy with a constant equation of state w and a specified sound speed c_s^2, allowing for perturbations in the dark energy component.
- `DarkEnergyPPF.f90`: Implements a Parametrized Post-Friedmann (PPF) dark en-

ergy perturbation model. For perturbations, it introduces an auxiliary variable Γ governed by the PPF parameter `c_Gamma_ppf` to parameterize the anisotropic stress, avoiding the need to specify a dark energy sound speed for perturbations.

- `DarkEnergyQuintessence.f90`: Implements the quintessence model, where dark energy is described by a scalar field with a given potential. It calculates the contributions of the scalar field to the total energy density, pressure, and perturbations.

Note that terms involving $1/a$ should be avoided when modifying the dark energy evolution, as they lead to divergence and numerical blow-up as $a \to 0$.

4.3 CAMB Fortran Perturbation Equations

CAMB adopts the **CDM synchronous gauge** ($v_{\rm c} = 0$) for cosmological perturbation calculations. The perturbation equations in this gauge were first derived in [26, Ma and Bertschinger], and are formally documented in the official CAMB notebook https://camb.readthedocs.io/en/latest/ScalEqs.html and technical note https://cosmologist.info/notes/CAMB.pdf.

Given the inconsistent notation for perturbation variables between theoretical references and the CAMB Fortran source code, we start from the fundamental equations in [26, Ma and Bertschinger] and explicitly demonstrate their numerical implementation in CAMB.

We adopt a clear notation convention throughout this section: variables with the subscript "s" (e.g., η_s, h_s) refer to quantities as defined in [26, Ma and Bertschinger], while unsubscripted variables (e.g., η, h) follow the conventions used in the official CAMB documentation (ScalEqs.html and CAMB.pdf).

The perturbed Friedmann-Robertson-Walker (FLRW) metric in [26, Ma and Bertschinger] takes the form

$$\mathrm{d}s^2 = a^2(\tau)\left[-\mathrm{d}\tau^2 + (\delta_{ij} + h_{ij})\mathrm{d}x^i\mathrm{d}x^j\right], \tag{4.3}$$

where h_{ij} denotes the metric perturbation, which can be decomposed into scalar, vector, and tensor modes. For scalar perturbations, the Fourier-space representation of h_{ij} is

$$h_{ij}(\mathbf{k},\tau) = \hat{k}_i\hat{k}_j h_s(\mathbf{k},\tau) + \left(\hat{k}_i\hat{k}_j - \frac{1}{3}\delta_{ij}\right)6\eta_s(\mathbf{k},\tau), \tag{4.4}$$

$$= \frac{1}{3}\delta_{ij}h_s(\mathbf{k},\tau) + \left(\hat{k}_i\hat{k}_j - \frac{1}{3}\delta_{ij}\right)\left[h_s(\mathbf{k},\tau) + 6\eta_s(\mathbf{k},\tau)\right], \tag{4.5}$$

where $h_s = h_{ii}$ (the trace of the metric perturbation) and η_s are the two independent scalar metric variables. Mapping this metric to the general perturbed FLRW form in

Eqs. (2.37) yields the following gauge transformation relations:

$$\boxed{\begin{aligned} &\phi = 0, \quad B = 0, && (4.6)\\ &\psi = -\frac{h_s}{6}, \quad E = -\frac{1}{2k^2}(h_s + 6\eta_s). && (4.7)\end{aligned}}$$

Substituting these relations into the Bardeen potentials (Eqs. (2.49)) gives

$$\begin{aligned}\Psi &\equiv \phi + \mathcal{H}(B - E') + (B - E')' = \frac{1}{2k^2}\left[h_s'' + 6\eta_s'' + \mathcal{H}(h_s' + 6\eta_s')\right], \\ \Phi &\equiv \psi + \frac{1}{3}\nabla^2 E - \mathcal{H}(B - E') = -\frac{h_s}{6} - \frac{k^2}{3}\left(-\frac{1}{2k^2}(h_s + 6\eta_s)\right) + \mathcal{H}\left(-\frac{1}{2k^2}(h_s' + 6\eta_s')\right) \\ &= \eta_s - \frac{\mathcal{H}}{2k^2}(h_s' + 6\eta_s'),\end{aligned} \tag{4.8}$$

which exactly reproduce Eqs. (18) in [26, Ma and Bertschinger].

The full set of scalar perturbation equations in the synchronous gauge are given by Eqs. (21a)–(21d) in [26, Ma and Bertschinger]:

$$\boxed{\begin{aligned} k^2\eta_s - \frac{1}{2}\mathcal{H}h_s' &= 4\pi Ga^2\delta T^{0\ \mathrm{Syn}}_{\ 0} = -4\pi Ga^2\delta\rho^{\mathrm{Syn}}, && (21a)\\ k^2\eta_s' &= 4\pi Ga^2\, ik^j\delta T^{0\ \mathrm{Syn}}_{\ j} = -4\pi Ga^2k^2(\bar\rho + \bar P)v^{\mathrm{Syn}}, && (21b)\\ h_s'' + 2\mathcal{H}h_s' - 2k^2\eta_s &= -8\pi Ga^2\delta T^{i\ \mathrm{Syn}}_{\ i} = -24\pi Ga^2\delta P^{\mathrm{Syn}}, && (21c)\\ h_s'' + 6\eta_s'' + 2\mathcal{H}(h_s' + 6\eta_s') - 2k^2\eta_s &= 24\pi Ga^2\left(\hat k^i\hat k_j - \frac{1}{3}\delta^i_{\ j}\right)\left(T^j_{\ i} - \frac{1}{3}\delta^j_{\ i}T^k_{\ k}\right) && (21d)\\ &= -16\pi Ga^2k^2\Pi. \end{aligned}}$$

One key advantage of the synchronous gauge is the direct algebraic relation between the metric perturbations h_s, η_s and the perturbed energy-momentum tensor $\delta T^\mu_{\ \nu}$. In contrast to the conformal Newtonian gauge — where time derivatives of the gravitational potentials cannot be solved algebraically from the Poisson and momentum constraints (Eqs. (2.56) and (2.58)) without invoking the second-order spatial Einstein equations — the synchronous gauge enables explicit algebraic solutions for the first-order time derivatives of the metric variables directly from the energy-momentum tensor components. Rearranging Eqs. (21a) and (21b) gives

$$h_s' = \frac{2}{\mathcal{H}}\left[k^2\eta_s - 4\pi Ga^2\delta T^{0\ \mathrm{Syn}}_{\ 0}\right], \tag{4.9}$$

$$\eta_s' = \frac{4\pi Ga^2}{k^2}(\bar\rho + \bar P)\left(ik^j\delta T^{0\ \mathrm{Syn}}_{\ j}\right). \tag{4.10}$$

A second major advantage is that the Boltzmann equations for collisionless species (photons, massless neutrinos, and massive neutrinos) contain no spatial derivatives of the gravitational potentials in this gauge. All metric perturbation terms depend solely on conformal time τ, with no explicit dependence on the comoving wavenumber k. This structure allows for vectorized numerical integration over all k-modes simultaneously, delivering a level of computational efficiency that is unattainable in other gauges, where metric terms introduce k-dependent spatial derivatives that require separate evaluation for each individual mode.

Nonetheless, the synchronous gauge admits residual gauge modes that can drive unphysical early-time growth and numerical instabilities. The standard gauge-fixing condition $v_{\mathrm{c}} = 0$ becomes ill-conditioned at very early epochs, when the CDM energy density is dynamically negligible. To suppress these spurious modes, CAMB implements a specialized treatment for the CDM component and applies the tight-coupling approximation for photons and baryons in the early Universe, ensuring stable and accurate evolution of cosmological perturbations across all cosmic times.

We now present the key variable definitions and equation mappings used in the CAMB source code.

`derivs` This subroutine computes the conformal time derivatives of scalar perturbation variables in `equations.f90`. The main definitions are as follows:

- **Background equations:**
  $\texttt{grho} \equiv 8\pi G a^2 \bar{\rho} = 8\pi G a^2 \sum_i \bar{\rho}_i = \texttt{grhocrit}\left(\frac{\Omega_{\mathrm{b0}}+\Omega_{\mathrm{c0}}}{a} + \frac{\Omega_{\mathrm{g0}}+\Omega_{\mathrm{r0}}}{a^2}\right) + \texttt{grho}_\nu + \texttt{grhov\_t}$. Note that this term excludes curvature contributions; the Hubble parameter used in perturbation calculations is

  $$\mathcal{H} \equiv \texttt{adotoa} = \sqrt{\frac{\texttt{grho} + \texttt{grhocrit}\cdot\Omega_{\mathrm{k0}}}{3}}. \qquad \text{(lines 2231–2237)}$$

  $\texttt{gpres} \equiv 8\pi G a^2 \bar{P} = 8\pi G a^2 \sum_i \bar{P}_i = \texttt{grhocrit}\left(\frac{\Omega_{\mathrm{g0}}+\Omega_{\mathrm{r0}}}{a^2}\right)\frac{1}{3} + \texttt{gpres}_\nu + \texttt{grhov\_t}\, w_{\mathrm{de}}(a)$ is the total background pressure, used to compute

  $$\texttt{adotdota} \equiv \frac{a''}{a} = \frac{1}{2}\left(\mathcal{H}^2 - 8\pi G a^2 \bar{P}\right) = (\texttt{adotoa} * \texttt{adotoa} - \texttt{gpres})/2,$$

  and the conformal time derivative of the Weyl potential $\texttt{phidot} \equiv (\Psi + \Phi)'/2$.

- **Perturbation equations:**
  For numerical stability and efficiency, `CAMB` rescales and non-dimensionalizes equations in the Fourier domain based on the formalism of ScalEqs.html, with all evolution equations solved in conformal time τ and comoving wavenumber k space. The core code variables and their evolution equations are algebraic transformations of Eqs. (21a)–(21b). The rescaled variables $\texttt{etak} \equiv \eta_s k$ and `z` are introduced to

remove the k-dependent numerical singularity and ensure stable evolution as $k \to 0$:

$$\begin{aligned}
\texttt{z} &= \frac{h_s'}{2k} = \frac{3h'}{k} = \frac{\texttt{dgrho} + 2\texttt{etak}\,k}{2k\mathcal{H}}, \quad \text{(line 2294)},\\
\texttt{sigma} \equiv \sigma &= \frac{h_s' + 6\eta_s'}{2k} = \texttt{z} + \frac{3}{2k^2}\texttt{dgq}, \quad \text{(line 2297)},\\
\texttt{ayprime(ix_etak)} \equiv \texttt{etak}' &= \frac{1}{2}\texttt{dgq} + \texttt{State\%curv}\,\texttt{z}, \quad \text{(line 2301)},\\
\texttt{phi} \equiv \frac{\Psi + \Phi}{2} &= -\frac{\texttt{dgrho} + \frac{3\texttt{dgq}\mathcal{H}}{k} + \texttt{dgpi}}{2k^2}, \quad \text{(line 2693)},
\end{aligned}$$

where

$$\begin{aligned}
\texttt{etak} &\equiv \eta_s k = -\frac{1}{2}\eta k,\\
\texttt{dgrho} &\equiv 8\pi G a^2 \delta\rho^{\rm Syn} = 8\pi G a^2 \sum_i \bar{\rho}_i \Delta_i,\\
\texttt{dgq} &\equiv 8\pi G a^2 (\bar{\rho} + \bar{P})(-v_s^{\rm Syn}) = 8\pi G a^2 q(t) = 8\pi G a^2 \sum_i (\bar{\rho}_i + \bar{P}_i) v_i,\\
\texttt{dgpi} &\equiv 8\pi G a^2 \Pi(t).
\end{aligned}$$

- **Output**
  The subroutine computes and returns the right-hand side of the evolution equations, stored in the derivative vector y' (`ayprime`), for the vector y (`ay`) of core dynamical variables. These include: `etak` (`etak`), $\Delta_{\rm c}$ (`clxc`), $\Delta_{\rm b}$ (`clxb`), $v_{\rm b}$ (`vb`), $\Delta_{\rm g}$ (`clxg`), $\Delta_{\rm r}$ (`clxr`), $q_{\rm g}$ (`qg`), $\pi_{\rm r}$ (`pir`), $\pi_{\rm g}$ (`pig`), $q_{\rm r}$ (`qr`), and dark energy variables.

  It also outputs transfer functions via `OutputTransfer` (lines 2755-2775), together with temperature source terms (including the Sachs-Wolfe effect, Integrated Sachs-Wolfe effect, Doppler effect, and quadrupole source term) and the E-mode polarization source.

`initial` This subroutine defines the scalar perturbation initial conditions for the early, radiation-dominated Universe. It initializes the state vector y for all species (CDM, baryons, photons, neutrinos, dark energy) and the synchronous gauge metric perturbation `etak` at a given conformal time τ and wavenumber k. It supports adiabatic and isocurvature modes, configures the tight-coupling approximation, and initializes consistent perturbations for massive neutrinos, matching the variable structure of `derivs`.

Vector and tensor perturbations Vector and tensor perturbations are implemented via initialization routines (`initialv`, `initialt`) and conformal-time evolution modules (`derivsv`, `derivst`). They describe vorticity modes and primordial gravitational waves, support both tight-coupling approximation and full multipole expansions for photons and neutrinos, compute polarization and CMB source terms including B-modes, and maintain consistent variable indexing and gauge setup with the scalar perturbation framework.

Some Notes for CAMB Modifications

- **Friedmann Equations Modification**: One of the robust and efficient ways to modify $\mathcal{H}(a)$ and $\mathcal{H}'(a)$ is to adjust the *effective dark energy density* and *equation of state parameters* within the `BackgroundDensityAndPressure` module. This ensures that the changes propagate self-consistently to all parts of the code that depend on the Friedmann equations.

- **Integer Division Pitfall**: In Fortran, dividing two integers with `/` truncates the fractional part, which can introduce critical numerical errors. For example, `4/3` evaluates to `1` instead of the mathematical value $1.\dot{3}$. Always use double-precision real division for numerical calculations, e.g., `4.0_dl/3.0_dl`, where `dl` denotes the double-precision real kind parameter.

- **Normalized Hubble Parameter**: Sometimes you may need the ratio $E^2 = H^2/H_0^2$, here show an example to get the quantity:

$$\texttt{dtauda} = \frac{\mathrm{d}\tau}{\mathrm{d}a} = \sqrt{\frac{3}{\texttt{grhocrit} \cdot E^2 a^4}} = \sqrt{\frac{c^2}{H_0^2 \cdot E^2 a^4}} = \frac{c}{a^2 H}, \tag{4.11}$$

$$E^2 = \frac{H^2}{H_0^2} = \left(\frac{c/1000}{a^2 H_0 \texttt{dtauda}} \right)^2 . \tag{4.12}$$

- **Numerical Stability**: For robust numerical evolution, when implementing new equations, ensure all terms are well-behaved at both early and late times. Avoid expressions that could lead to division by zero or numerical overflow.

4.4 CAMB Fortran Modifications

This section presents practical guidelines for modifying the CAMB Fortran code to implement custom cosmological models.

Add parameters There are two ways to add new parameters for your cosmological model:

- Add parameters to the `CAMBparams` type defined in `fortran/model.f90`, along with the corresponding routines in `camb/model.py`—in the same locations as standard parameters like `H0`, `omch2`, and `ombh2`. This ensures the parameters are accessible across all source files and can be used identically to standard cosmological parameters.

- Add parameters to the specific dark energy `.f90` files in the `fortran` directory and to `camb/dark_energy.py`—for example, the dark energy parameters `w0` and `wa`. Note that such parameters are only accessible within the corresponding dark energy source files.

Modify code Modify the relevant equations to match your new cosmological model, typically by updating the corresponding components in the dark energy source files and utilizing the utility functions provided in `forutils`. It is critical to ensure the stability of the evolution equations and maintain full consistency with CAMB's internal framework.

Compile After implementing all code modifications, compile the updated CAMB in the terminal using the following commands:

```
cd /path/to/the/Modified_CAMB
python setup.py make
```

Once compiled successfully, the modified CAMB is ready for cosmological simulations and calculations!

4.5 CAMB Python Interface

The CAMB Python interface provides a high-level, user-friendly wrapper for the core Fortran cosmological Boltzmann solver. The official demonstration notebook https://camb.readthedocs.io/en/latest/CAMBdemo.html provides a comprehensive suite of executable workflows and visualization examples. For the detailed definitions and usages of specific output functions, users may refer to the source code in the `camb/results.py` module.

4.6 MCMC Sampling, Result Analysis and Visualization

With the ability to compute theoretical cosmological observables using `CAMB` (or its modified versions), we can leverage `Cobaya` , a Bayesian inference framework, to interface these theoretical predictions with observational datasets. Detailed installation instructions for `Cobaya` are provided in Section 1.2.

For the statistical analysis and visualization of posterior sampling results, `GetDist` serves as a specialized Python package tailored for processing Monte Carlo samples.

For the full functionality of these packages, we refer users to their official documentation at https://cobaya.readthedocs.io/en/latest/, https://getdist.readthedocs.

io/en/latest/, and the original source code in the packages.

To make your cosmological inference journey easier, we have also put together a collection of example posterior chains and executable Jupyter notebooks in the GitHub repository: https://github.com/XueZHOU-cosmology/Tutorial.

Appendix A

Command Cheat Sheet

Command	Core Function
`camb -version`	Check the installed version of CAMB
`python setup.py make`	Compile the CAMB Fortran backend
`cobaya -version`	Check the installed version of Cobaya
`cobaya cosmo-generator`	Launch the interactive generator for cosmological analysis yaml or python configuration files
`cobaya-run [config.yaml]`	Run a full Cobaya Bayesian sampling analysis from a yaml configuration file
`cobaya-run -r [config.yaml]`	Resume an interrupted Cobaya sampling run from the last checkpoint
`cobaya-install [likelihood_name]`	Install a specific Cobaya likelihood dataset and its corresponding data
`cobaya-install cosmo /packages/path`	One-click installation of cosmological theory codes (CAMB/CLASS) and standard likelihood datasets

Appendix B

Useful Resources and Community Links

- **CAMB Official Core Resources**
  - CAMB Official Repository: https://github.com/cmbant/CAMB
  - CAMB Official Homepage: https://camb.info/
  - CAMB Official Perturbation Equations Notebook: https://camb.readthedocs.io/en/latest/ScalEqs.html
  - CAMB Interactive Python Example Notebook: https://camb.readthedocs.io/en/latest/CAMBdemo.html
  - CAMB Notes : https://cosmologist.info/notes/CAMB.pdf
  - CosmoCoffee Forum (Cosmology Code & Analysis Community): https://cosmocoffee.info/
- **Cobaya & GetDist Official Core Resources**
  - Cobaya Official Repository: https://github.com/CobayaSampler/cobaya
  - Cobaya Official Documentation: https://cobaya.readthedocs.io
  - Cobaya Cosmological Basic Runs Tutorial: https://cobaya.readthedocs.io/en/latest/cosmo_basic_runs.html
  - Cobaya Likelihoods Installation Guide: https://cobaya.readthedocs.io/en/latest/installation_cosmo.html
  - Cobaya GUI Generator: https://cobaya-gui.streamlit.app/
  - GetDist Official Repository: https://github.com/cmbant/getdist
  - GetDist Official Documentation: https://getdist.readthedocs.io

  - GetDist Plot Gallery & Tutorial: https://getdist.readthedocs.io/en/latest/plot_gallery.html

- **Cosmological Observational Data & Likelihood Libraries**
  - Planck Legacy Archive (Planck CMB Data & Likelihoods): https://pla.esac.esa.int/pla/
  - SPT-3G CMB Data & Likelihoods (candl Framework): https://github.com/SouthPoleTelescop spt_candl_data
  - Pantheon+ & SH0ES Type Ia Supernovae Data Release: https://github.com/PantheonPlusSH0ES/DataRelease
  - DESI BAO/RSD Data & Likelihoods: https://desi.lbl.gov/science/cosmology-results/bao/

Bibliography

[1] Erik Rosenberg, Steven Gratton, and George Efstathiou. "CMB power spectra and cosmological parameters from Planck PR4 with CamSpec". In: *Mon. Not. Roy. Astron. Soc.* 517.3 (2022), pp. 4620–4636. DOI: 10.1093/mnras/stac2744. arXiv: 2205.10869 [astro-ph.CO].

[2] David W. Hogg et al. "Cosmic homogeneity demonstrated with luminous red galaxies". In: *Astrophys. J.* 624 (2005), pp. 54–58. DOI: 10.1086/429084. arXiv: astro-ph/0411197.

[3] Pavan Kumar Aluri et al. "Is the observable Universe consistent with the cosmological principle?" In: *Class. Quant. Grav.* 40.9 (2023), p. 094001. DOI: 10.1088/1361-6382/acbefc. arXiv: 2207.05765 [astro-ph.CO].

[4] Rodrigo S. Gonçalves et al. "Measuring the cosmic homogeneity scale with SDSS-IV DR16 Quasars". In: *JCAP* 03 (2021), p. 029. DOI: 10.1088/1475-7516/2021/03/029. arXiv: 2010.06635 [astro-ph.CO].

[5] G J Whitrow. *Gravitation and Cosmology: Principles and Applications of the General Theory of Relativity.* Vol. 25. 2. 1974, pp. 65–65.

[6] John A. Peacock. *Cosmological Physics.* Cambridge: Cambridge University Press, 1999.

[7] Charles W. Misner, K. S. Thorne, and J. A. Wheeler. *Gravitation.* San Francisco: W. H. Freeman, 1973. ISBN: 978-0-691-17779-3.

[8] Patrick Peter and Jean-Philippe Uzan. *Primordial Cosmology.* Oxford Graduate Texts. Oxford University Press, Feb. 2013. ISBN: 978-0-19-920991-0.

[9] N. Straumann. "Minimal assumptions leading to a Robertson-Walker model of the universe". In: *Helvetica Physica Acta* 47.3 (1974), pp. 879–880. DOI: 10.5169/seals-114576.

[10] Stephen W. Hawking and George F. R. Ellis. *The Large Scale Structure of Space-Time.* Cambridge Monographs on Mathematical Physics. Cambridge University Press, Feb. 2023. ISBN: 978-1-009-25316-1. DOI: 10.1017/9781009253161.

[11] Robert M. Wald. *General Relativity*. Chicago, USA: Chicago Univ. Pr., 1984. DOI: 10.7208/chicago/9780226870373.001.0001.

[12] Lavinia Heisenberg. “A Re-Examination Of Foundational Elements Of Cosmology”. In: (Dec. 2025). arXiv: 2512.16934 [physics.gen-ph].

[13] James M. Bardeen. “Gauge-invariant cosmological perturbations”. In: *Phys. Rev. D* 22 (8 1980), pp. 1882–1905. DOI: 10.1103/PhysRevD.22.1882. URL: https://link.aps.org/doi/10.1103/PhysRevD.22.1882.

[14] E. Lifshitz. “Republication of: On the gravitational stability of the expanding universe”. In: *J. Phys. (USSR)* 10.2 (1946), p. 116. DOI: 10.1007/s10714-016-2165-8.

[15] Juan Maldacena. “Non-gaussian features of primordial fluctuations in single field inflationary models”. In: *Journal of High Energy Physics* 2003.05 (2003), p. 013. DOI: 10.1088/1126-6708/2003/05/013. URL: https://doi.org/10.1088/1126-6708/2003/05/013.

[16] Erminia Calabrese et al. “The Atacama Cosmology Telescope: DR6 constraints on extended cosmological models”. In: *JCAP* 11 (2025), p. 063. DOI: 10.1088/1475-7516/2025/11/063. arXiv: 2503.14454 [astro-ph.CO].

[17] Y. Akrami et al. “Planck 2018 results. X. Constraints on inflation”. In: *Astron. Astrophys.* 641 (2020), A10. DOI: 10.1051/0004-6361/201833887. arXiv: 1807.06211 [astro-ph.CO].

[18] N. Aghanim et al. “Planck 2018 results. VI. Cosmological parameters”. In: *Astron. Astrophys.* 641 (2020), A6. DOI: 10.1051/0004-6361/201833910. arXiv: 1807.06209 [astro-ph.CO].

[19] Daniel Baumann. *Cosmology*. Cambridge University Press, July 2022. ISBN: 978-1-108-83807-8. DOI: 10.1017/9781108937092.

[20] R. K. Sachs and A. M. Wolfe. “Perturbations of a Cosmological Model and Angular Variations of the Microwave Background”. In: *Astrophys. J.* 147 (1967), p. 73. DOI: 10.1086/148961.

[21] P. A. R. Ade et al. “Planck 2013 results. XIX. The integrated Sachs-Wolfe effect”. In: *Astron. Astrophys.* 571 (2014), A19. DOI: 10.1051/0004-6361/201321526. arXiv: 1303.5079 [astro-ph.CO].

[22] P. A. R. Ade et al. “Planck 2015 results. XXI. The integrated Sachs-Wolfe effect”. In: *Astron. Astrophys.* 594 (2016), A21. DOI: 10.1051/0004-6361/201525831. arXiv: 1502.01595 [astro-ph.CO].

[23] Julien Carron, Antony Lewis, and Giulio Fabbian. “Planck integrated Sachs-Wolfe-lensing likelihood and the CMB temperature”. In: *Phys. Rev. D* 106.10 (2022), p. 103507. DOI: 10.1103/PhysRevD.106.103507. arXiv: 2209.07395 [astro-ph.CO].

[24] Wayne Hu and Naoshi Sugiyama. “Anisotropies in the cosmic microwave background: An Analytic approach”. In: *Astrophys. J.* 444 (1995), pp. 489–506. DOI: 10.1086/175624. arXiv: astro-ph/9407093.

[25] Wayne Hu and Naoshi Sugiyama. “Small scale cosmological perturbations: An Analytic approach”. In: *Astrophys. J.* 471 (1996), pp. 542–570. DOI: 10.1086/177989. arXiv: astro-ph/9510117.

[26] Chung-Pei Ma and Edmund Bertschinger. “Cosmological perturbation theory in the synchronous and conformal Newtonian gauges”. In: *Astrophys. J.* 455 (1995), pp. 7–25. DOI: 10.1086/176550. arXiv: astro-ph/9506072.

[27] Sunny Vagnozzi. “Consistency tests of LCDM from the early integrated Sachs-Wolfe effect: Implications for early-time new physics and the Hubble tension”. In: *Phys. Rev. D* 104.6 (2021), p. 063524. DOI: 10.1103/PhysRevD.104.063524. arXiv: 2105.10425 [astro-ph.CO].

[28] J. Colin Hill et al. “Atacama Cosmology Telescope: Constraints on prerecombination early dark energy”. In: *Phys. Rev. D* 105.12 (2022), p. 123536. DOI: 10.1103/PhysRevD.105.123536. arXiv: 2109.04451 [astro-ph.CO].

[29] Bjoern Malte Schaefer. “The integrated Sachs-Wolfe effect in cosmologies with coupled dark matter and dark energy”. In: *Mon. Not. Roy. Astron. Soc.* 388 (2008), pp. 1403–1408. DOI: 10.1111/j.1365-2966.2008.13521.x. arXiv: 0803.2239 [astro-ph].

[30] Emeric Seraille, Johannes Noller, and Blake D. Sherwin. “Constraining dark energy with the integrated Sachs-Wolfe effect”. In: *Phys. Rev. D* 110.12 (2024), p. 123525. DOI: 10.1103/PhysRevD.110.123525. arXiv: 2401.06221 [astro-ph.CO].

[31] Anton Chudaykin, Martin Kunz, and Julien Carron. “Modified gravity constraints with the Planck ISW-lensing bispectrum”. In: *Phys. Rev. D* 112.8 (2025), p. 083537. DOI: 10.1103/9zjh-8htv. arXiv: 2503.09893 [astro-ph.CO].

[32] Julien Carron, Mark Mirmelstein, and Antony Lewis. “CMB lensing from Planck PR4 maps”. In: *JCAP* 09 (2022), p. 039. DOI: 10.1088/1475-7516/2022/09/039. arXiv: 2206.07773 [astro-ph.CO].

[33] N. Aghanim et al. “Planck 2018 results. VIII. Gravitational lensing”. In: *Astron. Astrophys.* 641 (2020), A8. DOI: 10.1051/0004-6361/201833886. arXiv: 1807.06210 [astro-ph.CO].

[34] Antony Lewis and Anthony Challinor. “Weak gravitational lensing of the CMB”. In: *Phys. Rept.* 429 (2006), pp. 1–65. DOI: 10.1016/j.physrep.2006.03.002. arXiv: astro-ph/0601594.

[35] J. A. Zebrowski et al. “SPT-3G D1: Constraints on inflationary gravitational waves with two years of SPT-3G data”. In: *Phys. Rev. D* 112.12 (2025), p. 123520. DOI: 10.1103/33wm-c6pt. arXiv: 2505.02827 [astro-ph.CO].

[36] Kevork Abazajian et al. “CMB-S4: Forecasting Constraints on Primordial Gravitational Waves”. In: *Astrophys. J.* 926.1 (2022), p. 54. DOI: 10.3847/1538-4357/ac1596. arXiv: 2008.12619 [astro-ph.CO].

[37] Sebastian Belkner et al. “CMB-S4: Iterative Internal Delensing and r Constraints”. In: *Astrophys. J.* 964.2 (2024), p. 148. DOI: 10.3847/1538-4357/ad2351. arXiv: 2310.06729 [astro-ph.CO].

[38] N. Kaiser. “Clustering in real space and in redshift space”. In: *Mon. Not. Roy. Astron. Soc.* 227 (1987), pp. 1–27. DOI: 10.1093/mnras/227.1.1.

[39] Shadab Alam et al. “Completed SDSS-IV extended Baryon Oscillation Spectroscopic Survey: Cosmological implications from two decades of spectroscopic surveys at the Apache Point Observatory”. In: *Phys. Rev. D* 103.8 (2021), p. 083533. DOI: 10.1103/PhysRevD.103.083533. arXiv: 2007.08991 [astro-ph.CO].

[40] Eric V. Linder. “Cosmic growth history and expansion history”. In: *Phys. Rev. D* 72 (2005), p. 043529. DOI: 10.1103/PhysRevD.72.043529. arXiv: astro-ph/0507263.

[41] A. G. Adame et al. “DESI 2024 VI: Cosmological Constraints from the Measurements of Baryon Acoustic Oscillations”. In: (Apr. 2024). arXiv: 2404.03002 [astro-ph.CO].

[42] A. G. Adame et al. “DESI 2024 III: Baryon Acoustic Oscillations from Galaxies and Quasars”. In: (Apr. 2024). arXiv: 2404.03000 [astro-ph.CO].

[43] A. G. Adame et al. “DESI 2024 IV: Baryon Acoustic Oscillations from the Lyman alpha forest”. In: *JCAP* 01 (2025), p. 124. DOI: 10.1088/1475-7516/2025/01/124. arXiv: 2404.03001 [astro-ph.CO].

[44] M. Abdul Karim et al. “DESI DR2 results. II. Measurements of baryon acoustic oscillations and cosmological constraints”. In: *Phys. Rev. D* 112.8 (2025), p. 083515. DOI: 10.1103/tr6y-kpc6. arXiv: 2503.14738 [astro-ph.CO].

[45] Dan Scolnic et al. “The Pantheon+ Analysis: The Full Dataset and Cosmological Constraints”. In: *Astrophys. J.* 938.2 (2022), p. 110. DOI: 10.3847/1538-4357/ac8c0a. arXiv: 2112.03863 [astro-ph.CO].

[46] M. M. Phillips. “The Absolute Magnitudes of Type IA Supernovae”. In: *Astrophys. J. Lett.* 413 (1993), p. L105. DOI: 10.1086/186970.

[47] Dillon Brout et al. “The Pantheon+ Analysis: Cosmological Constraints”. In: *Astrophys. J.* 938.2 (2022), p. 110. DOI: 10.3847/1538-4357/ac8e04. arXiv: 2202.04077 [astro-ph.CO].